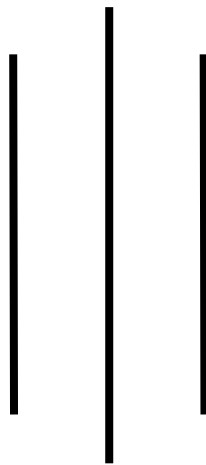


TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR



DATABASE ADMINISTRATION (BIT 352)

SUBMITTED BY:

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1. EXPLORING THE MULTITENANT ARCHITECTURE OF ORACLE 19C.

THEORY:

Oracle Database 19c uses a **Multitenant Architecture**, where a single **Container Database (CDB)** can contain multiple **Pluggable Databases (PDBs)**. This architecture simplifies database administration, improves resource utilization, and makes backup, recovery, and upgrades easier.

The main components are:

- **CDB (Container Database):** The main database that contains the Oracle system metadata and one or more PDBs.
- **PDB (Pluggable Database):** A portable database that stores user data and applications. Each PDB operates like an independent database.
- **Root Container (CDB\$ROOT):** Stores Oracle system Metadata and manages all PDBs.
- **PDB\$SEED:** A template database used to create new PDBs.
- **OPEN MODE**

Open Mode indicates the current state of a Container Database (CDB) or Pluggable Database (PDB). It determines whether the database is available for users to access and perform operations.

Common open modes are:

- **READ WRITE** – Users can read and modify data.
- **READ ONLY** – Users can only read data; modifications are not allowed.
- **MOUNTED** – The database is mounted but not open for normal user access.
- **ALTER SESSION SET CONTAINER:** Used to switch the current session between the root container and a pluggable database.

STEPS:

1. **Display current container:**
SHOW CON_NAME;
2. **Display information about the Container Database**
SELECT NAME,
CDB,
OPEN_MODE
FROM V\$DATABASE;
3. **Display all Pluggable Databases**
SHOW PDBS;
4. **Switch from the Root Container to a Pluggable Database**
ALTER SESSION SET CONTAINER = ORCLPDB;

5. Verify the current container

```
SHOW CON_NAME;
```

6. Return to the Root Container

```
ALTER SESSION SET CONTAINER = CDB$ROOT;
```

```
SHOW CON_NAME;
```

7. View the open mode of all PDBs

```
SHOW PDBS;
```

8. Change the open mode of a PDB

- **Close the PDB:**

```
ALTER PLUGGABLE DATABASE ORCLPDB CLOSE IMMEDIATE;
```

- **Check status:**

```
SHOW PDBS;
```

- **Open it again:**

```
ALTER PLUGGABLE DATABASE ORCLPDB OPEN;
```

- **Verify:**

```
SHOW PDBS;
```

OUTPUT:

1. Display current container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

2. Display information about the Container Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select name, cdb, open_mode from v$database;

NAME          CDB OPEN_MODE
-----
ORCL          YES READ WRITE
```

3. Display all Pluggable Databases

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show pdbs;

  CON_ID CON_NAME                                OPEN MODE RESTRICTED
-----
      2 PDB$SEED                                READ ONLY NO
      3 ORCLPDB                                READ WRITE NO
```

4. Switch from the Root Container to a Pluggable Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = orclpdb;
Session altered.
```

5. Verify the current container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
ORCLPDB
```

6. Return to the Root Container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = cdb$root;
Session altered.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

7. View the open mode of all PDBs

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
2	PDB\$SEED	READ ONLY	NO
3	ORCLPDB	READ WRITE	NO

8. Change the open mode of a PDB

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter pluggable database orclpdb close immediate;

Pluggable database altered.

SQL> show pdbs;

  CON_ID  CON_NAME                                OPEN  MODE RESTRICTED
-----
      2  PDB$SEED                                READ  ONLY      NO
      3  ORCLPDB                                MOUNTED
SQL> alter pluggable database orclpdb open;

Pluggable database altered.

SQL> show pdbs;

  CON_ID  CON_NAME                                OPEN  MODE RESTRICTED
-----
      2  PDB$SEED                                READ  ONLY      NO
      3  ORCLPDB                                READ  WRITE     NO
```

2. WORKING WITH STARTUP AND SHUTDOWN MODES IN ORACLE.

THEORY:

Oracle Database provides different **startup** and **shutdown** modes to control how the database is started and stopped. These modes are used for database administration, maintenance, recovery, and backup.

Startup Modes

- **STARTUP NOMOUNT** – Starts the Oracle instance by allocating memory and starting background processes. The control files are not opened.
- **STARTUP MOUNT** – Opens the control files but does not open the database. Used for recovery and maintenance operations.
- **STARTUP OPEN** (or simply **STARTUP**) – Opens the database for normal user access.
- **STARTUP FORCE** – Shuts down the currently running database (if necessary) and restarts it.
- **ALTER SYSTEM SUSPEND** – Suspends all database I/O operations temporarily.
- **ALTER SYSTEM RESUME** – Resumes normal database operations after suspension.

Shutdown Modes

- **SHUTDOWN NORMAL** – Waits for all users to disconnect before shutting down the database. No new connections are allowed.
- **SHUTDOWN TRANSACTIONAL** – Waits for active transactions to complete before shutting down.
- **SHUTDOWN IMMEDIATE** – Rolls back uncommitted transactions, disconnects users, and shuts down the database immediately.
- **SHUTDOWN ABORT** – Terminates the database immediately without waiting for users or transactions. Recovery is performed automatically during the next startup.

STEPS:

STARTUP MODES

1. Startup Database Normally

STARTUP;

2. Startup Database in NOMOUNT Mode

SHUTDOWN IMMEDIATE;

STARTUP NOMOUNT;

3. Startup Database in MOUNT Mode

SHUTDOWN IMMEDIATE;

STARTUP MOUNT;

4. Open the Database

ALTER DATABASE OPEN;

5. Startup Force

STARTUP FORCE;

6. Suspend Database

ALTER SYSTEM SUSPEND;

7. Resume Database

ALTER SYSTEM RESUME;

SHUTDOWN MODES

1. Shutdown Normal

SHUTDOWN NORMAL;

2. Shutdown Transactional

SHUTDOWN TRANSACTIONAL;

3. Shutdown Immediate

SHUTDOWN IMMEDIATE;

4. Shutdown Abort

SHUTDOWN ABORT;

OUTPUT:

STARTUP MODES:

1. Startup database normally

```
SQL> startup;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
Database opened.
```

2. Startup database in NOMOUNT mode

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup nomount;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
```

3. Startup database in MOUNT mode

```
SQL> startup mount;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
```

4. Open the database

```
SQL> alter database open;

Database altered.

SQL> set serveroutput on;
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

5. Startup force

```
SQL> startup force;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                 9277960 bytes
Variable Size             989855744 bytes
Database Buffers          3909091328 bytes
Redo Buffers               7495680 bytes
Database mounted.
Database opened.
SQL> exec my_info;

PL/SQL procedure successfully completed.

SQL> set serveroutput on;
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

6. Suspend database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter system suspend;

System altered.
```

7. Resume database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter system resume;

System altered.
```

SHUTDOWN MODES:

1. Shutdown normal

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown normal;
Database closed.
Database dismounted.
ORACLE instance shut down.
```

2. Shutdown transactional

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown transactional
Database closed.
Database dismounted.
ORACLE instance shut down.
```

3. Shutdown immediate

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

4. Shutdown abort

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown abort
ORACLE instance shut down.
```

3. USING ORACLE NET CONFIGURATION ASSISTANT (NETCA) FOR NET SERVICE CONFIGURATION.

THEORY:

Oracle Net

Oracle Net is Oracle's networking technology that enables communication between Oracle clients and Oracle database servers over a network.

Oracle Listener

An **Oracle Listener** is a server-side process that listens for incoming client connection requests and establishes connections between Oracle clients and the database. By default, the listener uses the **TCP protocol** on **port 1521**.

TNS (Transparent Network Substrate)

TNS is Oracle's networking protocol that enables communication between Oracle clients and Oracle databases. It allows users to connect to a database using a service name instead of specifying the complete connection details.

Example:

CONNECT system/password@ORCLPDB

listener.ora

The **listener.ora** file stores the configuration of Oracle Listeners, including the listener name, protocol, host, port number, and database services. It is automatically updated by NetCA.

Default location:

ORACLE_HOME\network\admin

tnsnames.ora

The **tnsnames.ora** file stores Local Net Service Names that map a service alias to the database connection details such as host, port, and service name. Oracle clients use this file to connect to the database.

Default location:

ORACLE_HOME\network\admin

Oracle Net Configuration Assistant (NetCA)

Oracle Net Configuration Assistant (NetCA) is a graphical utility used to configure Oracle Net Services. It helps users create, modify, test, and delete listeners and Local Net Service Names without manually editing the network configuration files.

STEPS:

Part A: Listener Configuration

Step 1: Add a Listener

1. *Select **Listener Configuration**.*
2. *Click **Next**.*
3. *Select **Add**.*
4. *Enter the Listener Name*
5. *Select **TCP** protocol.*
6. *Enter the Port Number (**1521**).*
7. *Click **Next**.*
8. *Finish the configuration.*

Step 2: Test the Listener

1. *Open **NetCA**.*
2. *Select **Listener Configuration**.*
3. *Verify that the listener has been successfully created.*
4. *Ensure the listener is active.*

Step 3: Delete the Listener

1. *Select **Listener Configuration**.*
2. *Select **Delete**.*
3. *Choose the Listener (**LISTENER**).*
4. *Click **Next**.*
5. *Finish the deletion.*

Part B: Local Net Service Name Configuration

Step 4: Add a Local Net Service Name

1. *Select **Local Net Service Name Configuration**.*

2. Click **Add**.
3. Enter the Service Name (e.g., **ORCLPDB**).
4. Select **TCP** as the protocol.
5. Enter the Host Name (e.g., **localhost**).
6. Enter the Listener Port (**1521**).
7. Enter the Username and Password.
8. Test the connection.
9. Assign a Net Service Name.
10. Finish the configuration.

Step 5: Test the Local Net Service Name

Use the **Test** option provided by NetCA.

OR

Verify using SQL*Plus.

CONNECT system/password@ORCLPDB;

Step 6: Delete the Local Net Service Name

1. Select **Local Net Service Name Configuration**.
2. Choose **Delete**.
3. Select the configured Net Service Name.
4. Finish the deletion.

Part C: Listener Control Commands

Step 7: Start the Listener

Open **Command Prompt** and execute:

lsnrctl start

Step 8: Check Listener Status

lsnrctl status

Step 9: Reload the Listener

lsnrctl reload

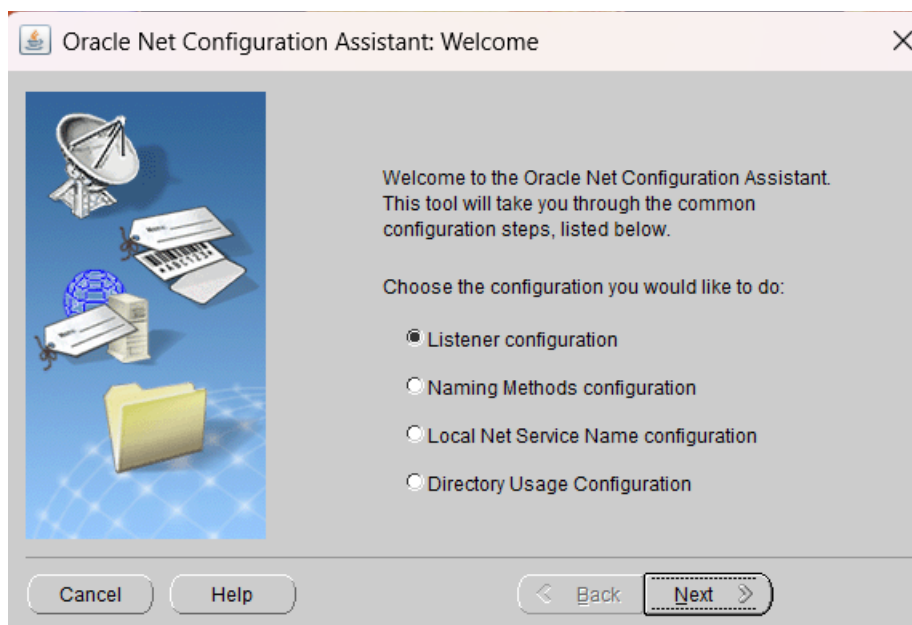
Step 10: Stop the Listener

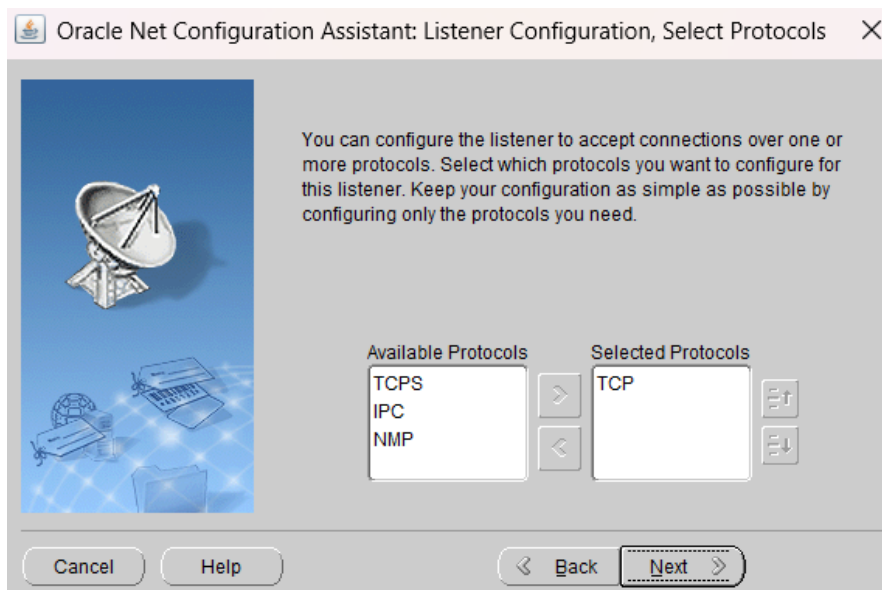
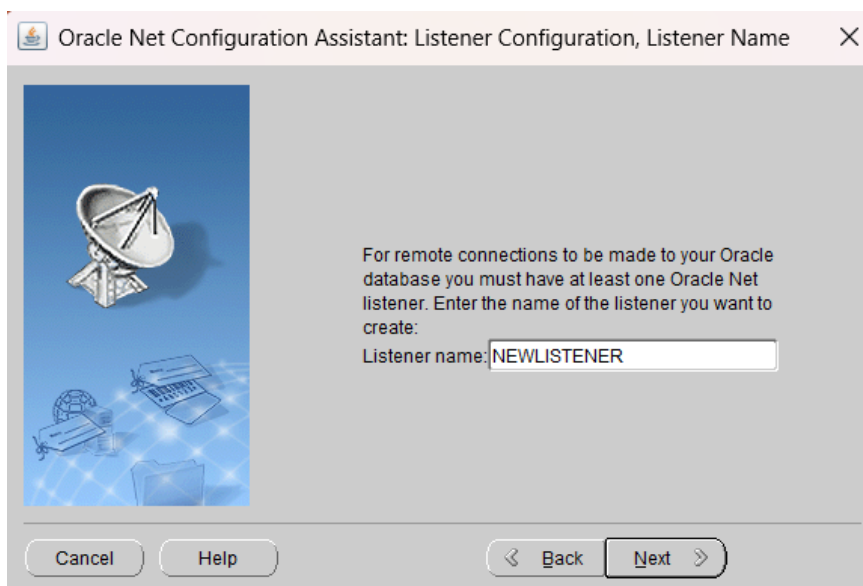
lsnrctl stop

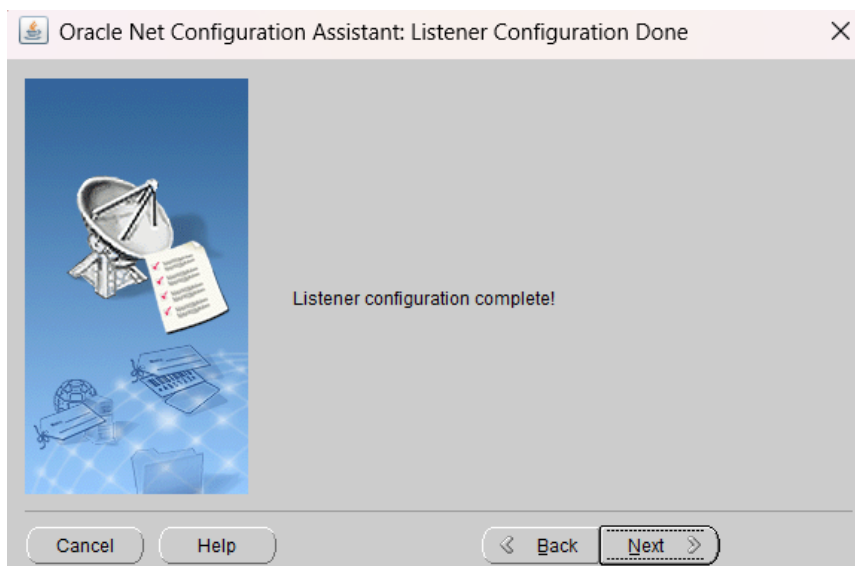
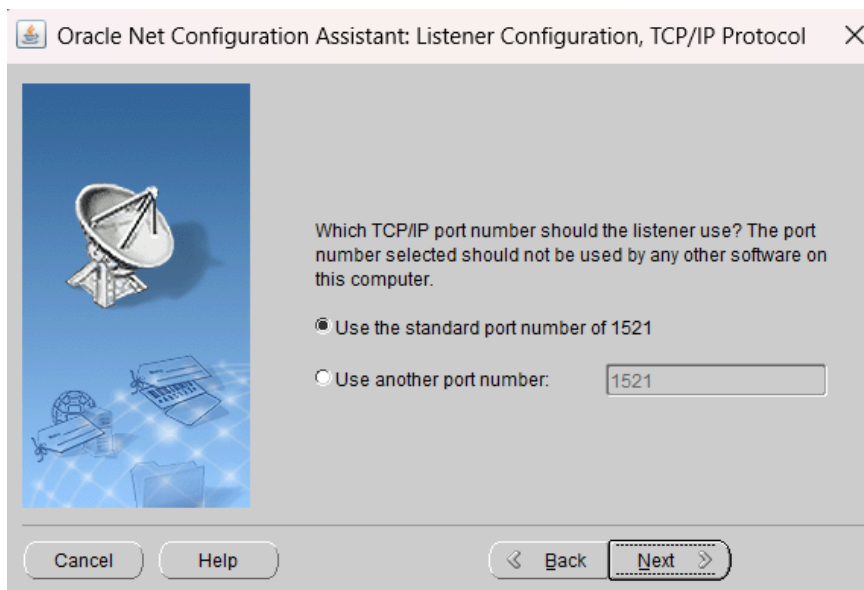
OUTPUT:

PART A: Listener Configuration

Step 1: Add a Listener







Step 2: Test the listener

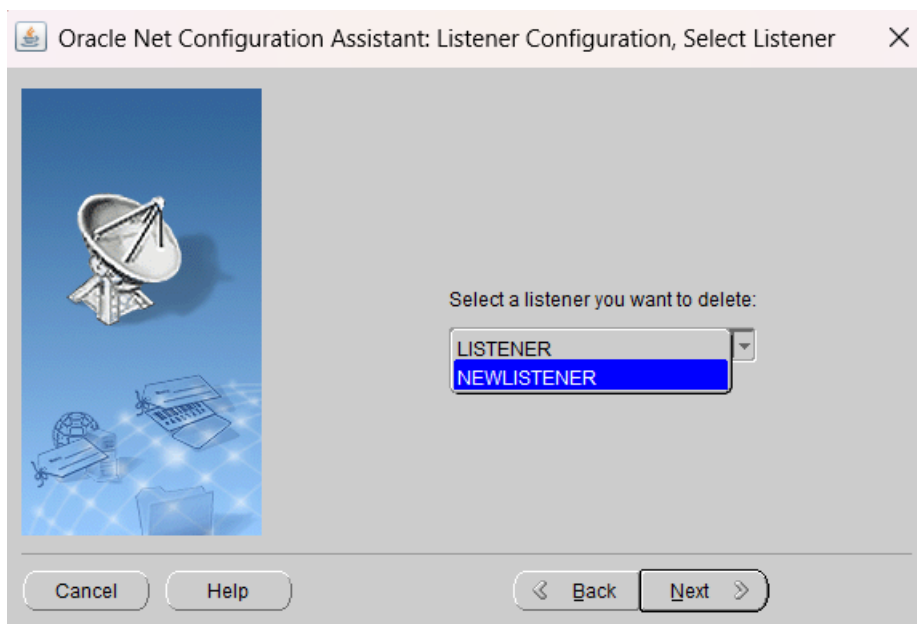
```
C:\Users\VICTUS>lsnrctl status NEWLISTENER

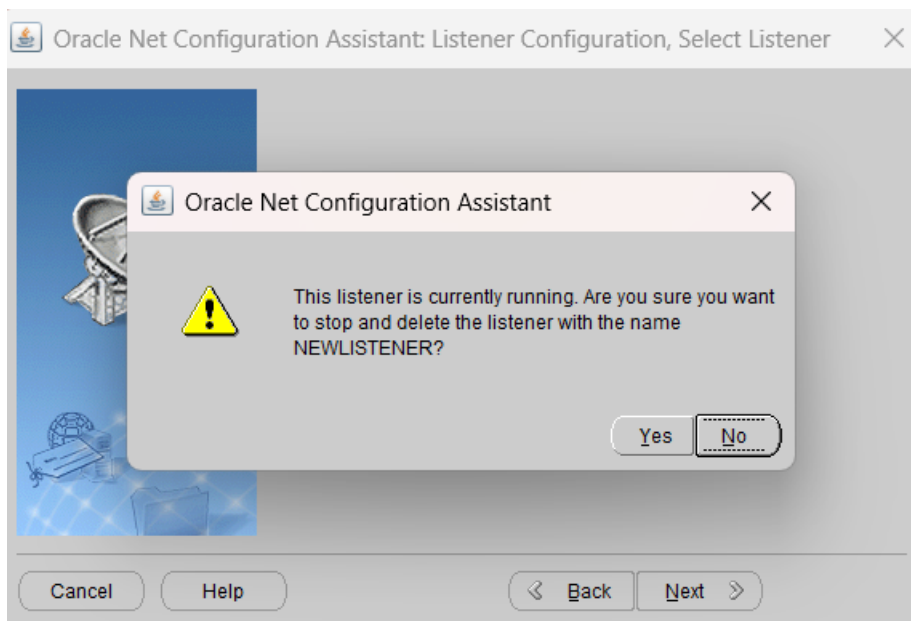
LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:00:55

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.1.118)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date                21-JUN-2026 08:45:07
Uptime                    21 days 6 hr. 15 min. 48 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   C:\Oracle_19C\network\admin\listener.ora
Listener Log File         C:\Oracle19C\diag\tnslnsr\Aconitin\listener>alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Aconitin)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
```

Step 3: Delete the listener





Verify whether the listener has been deleted

```
C:\Users\VICTUS\suresh>lsnrctl status NEWLISTENER

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:06:03

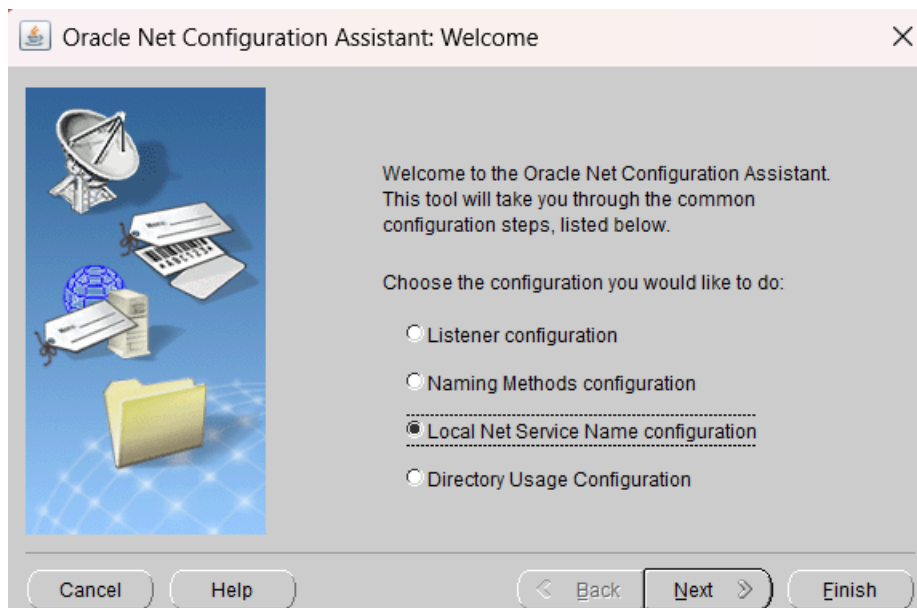
Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01101: Could not find listener name or service name NEWLISTENER
```

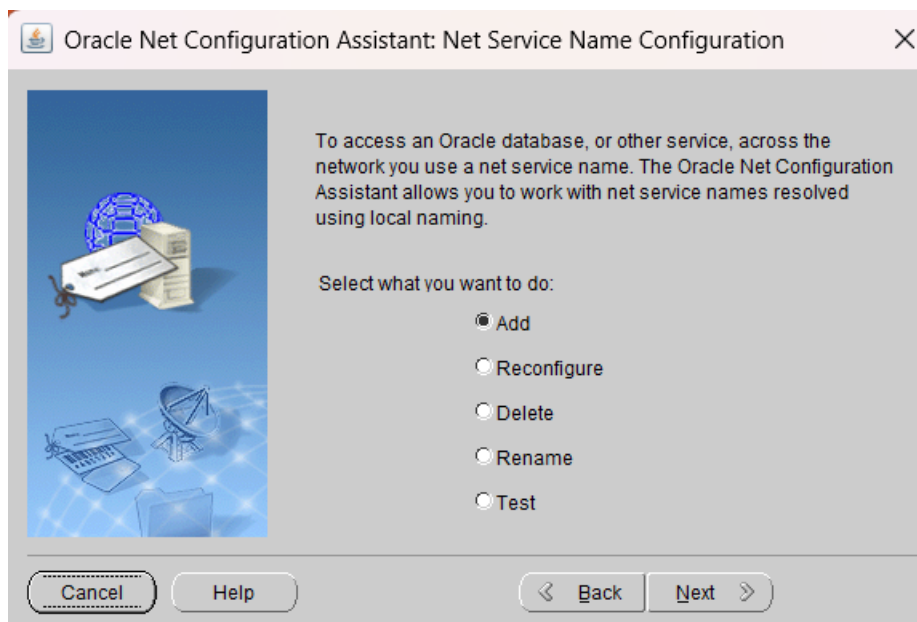
Part B: Local Net Service Name Configuration

Step 4: Add a Local Net Service Name

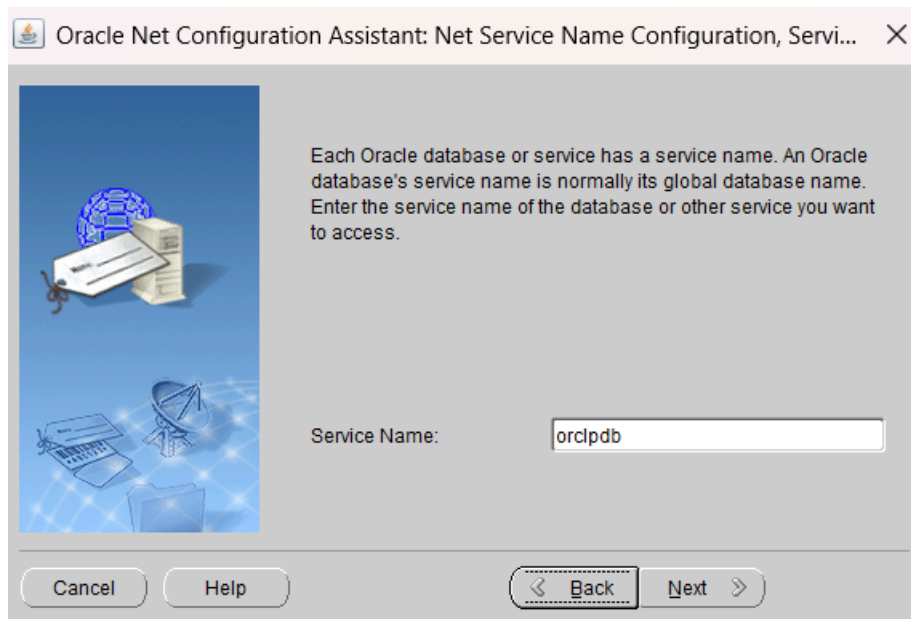
11. Select *Local Net Service Name Configuration*.



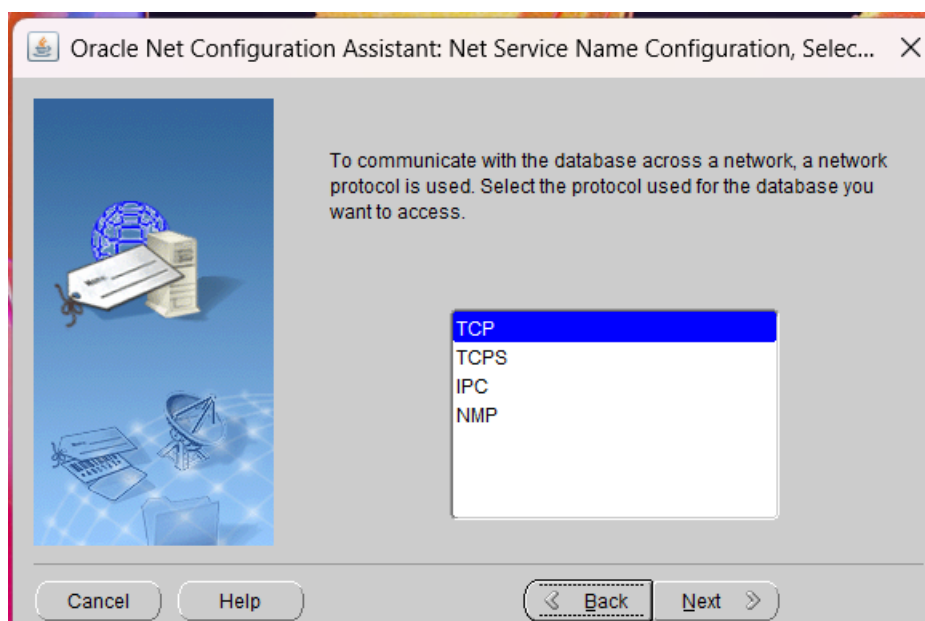
12. Click **Add**.



13. Enter the Service Name (e.g., **ORCLPDB**).



14. Select **TCP** as the protocol.



15. Enter the Host Name (e.g., **localhost**).

Oracle Net Configuration Assistant: Net Service Name Configuration, TCP/...



To communicate with the database using the TCP/IP protocol, the database computer's host name is required. Enter the host name for the computer where the database is located.

Host name:

A TCP/IP port number is also required. In most cases the standard port number should be used.

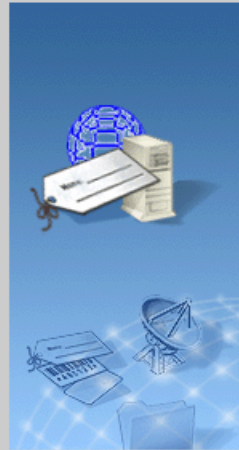
☒ Use the standard port number of 1521

☐ Use another port number:

Cancel Help Back Next

16. *Test the connection.*

Oracle Net Configuration Assistant: Net Service Name Configuration, Test



You can verify that an Oracle database can be reached, using the information provided, by performing a connection test.

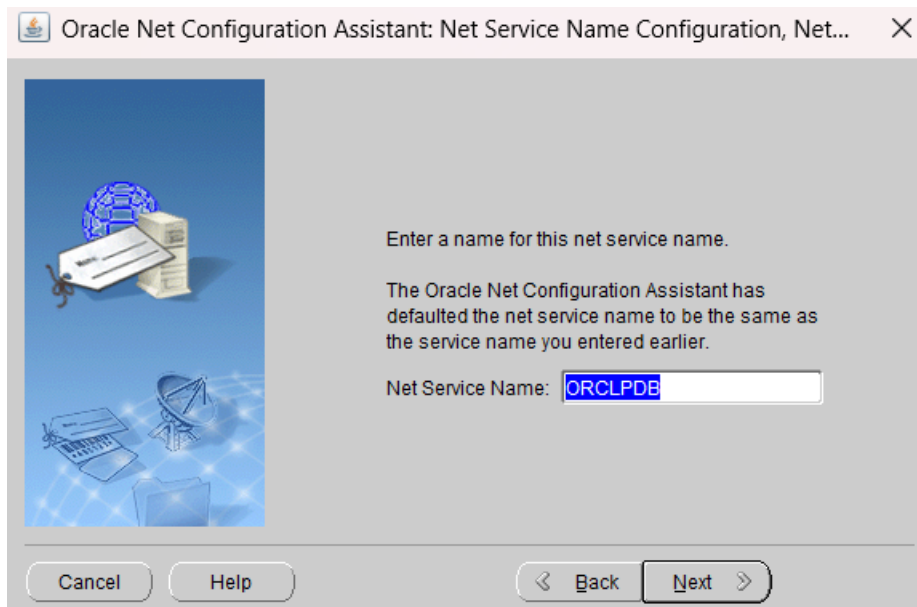
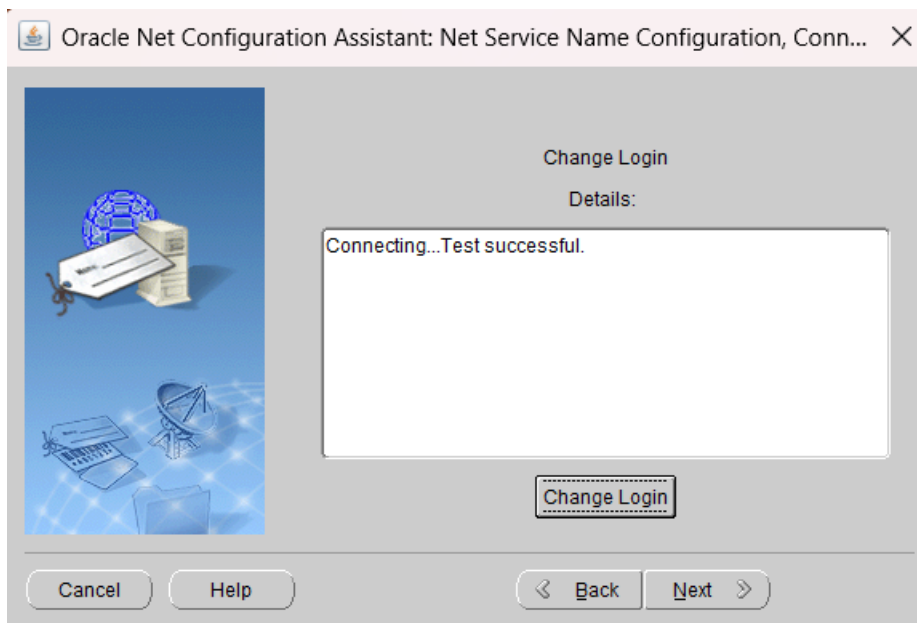
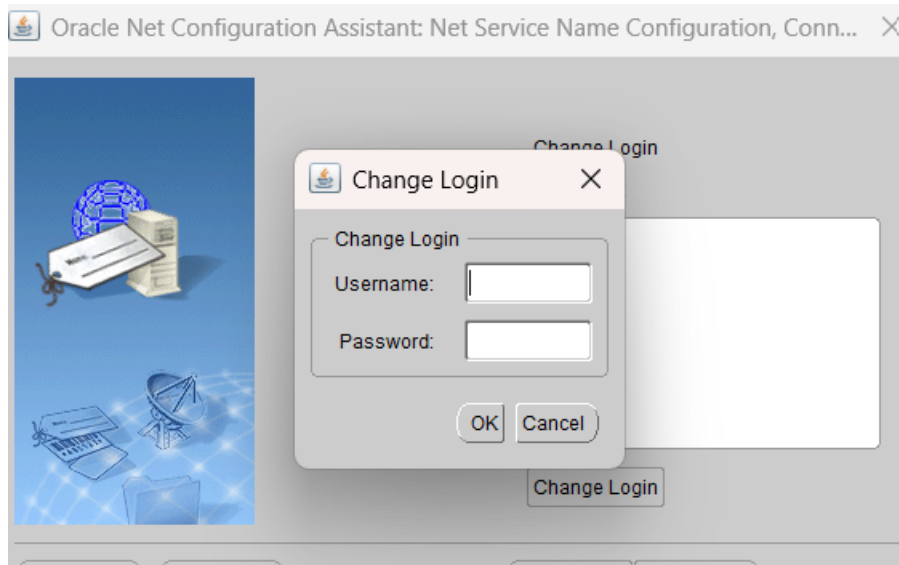
Would you like to test that a connection can be made to the database?

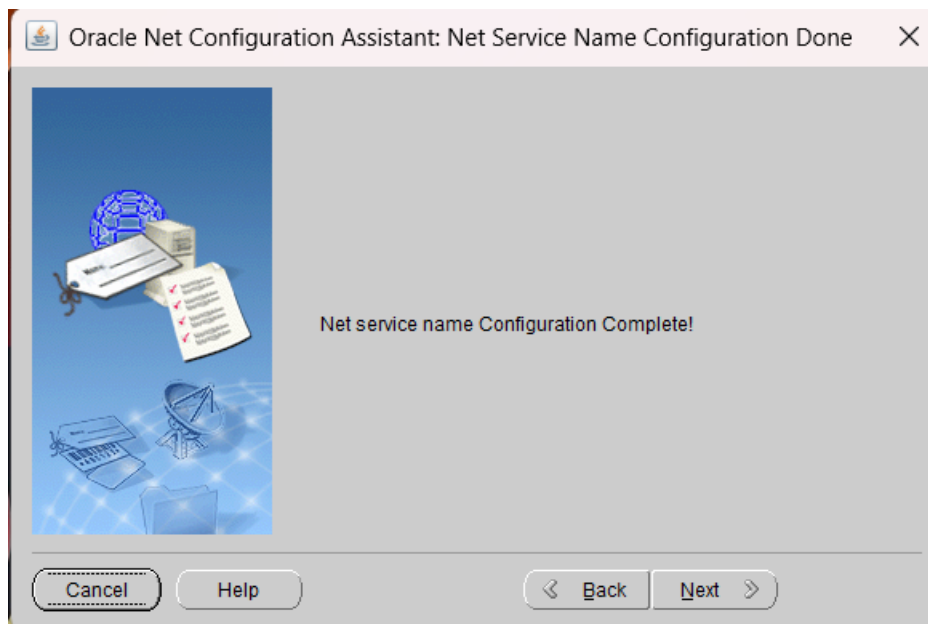
☐ No, do not test

☒ Yes, perform a test

Cancel Help Back Next

17. *Change login username and password*

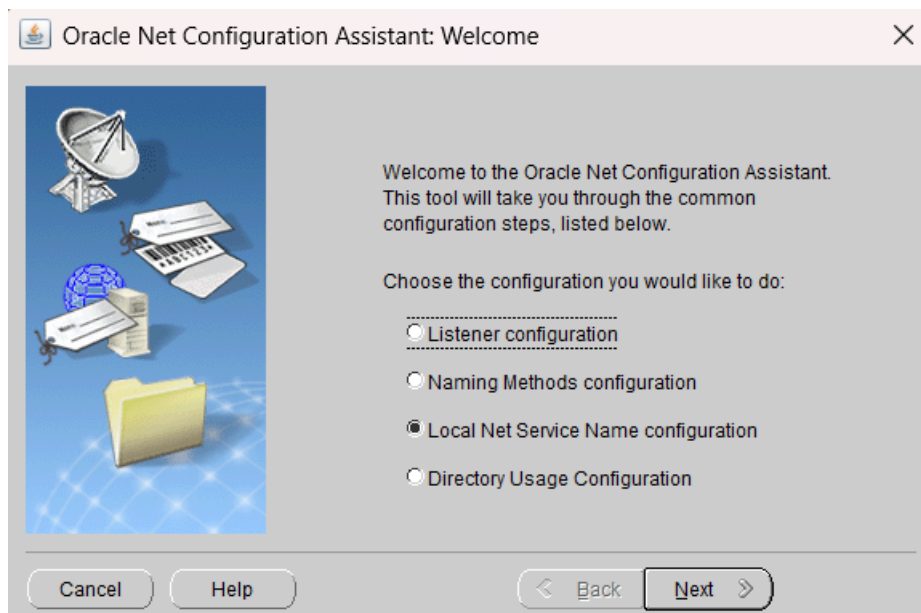


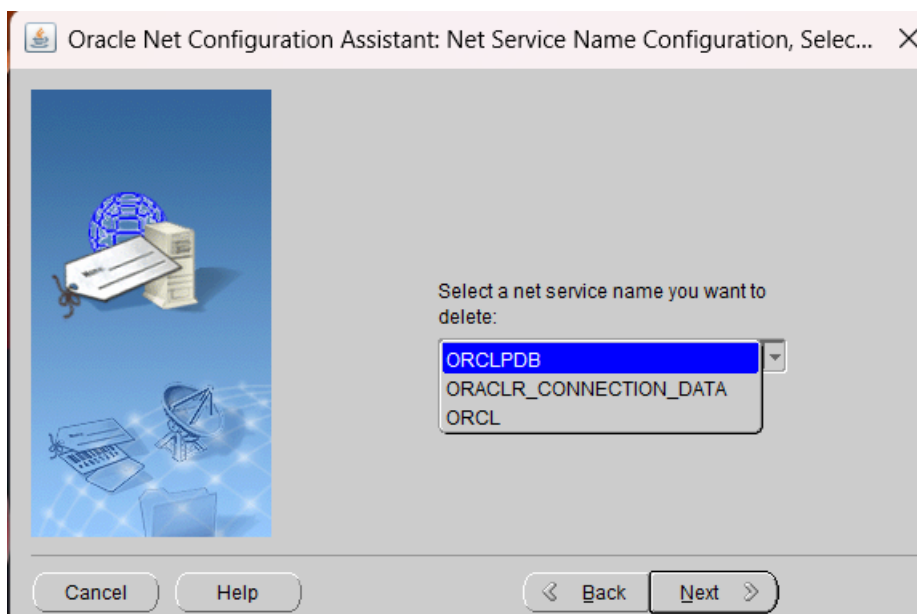
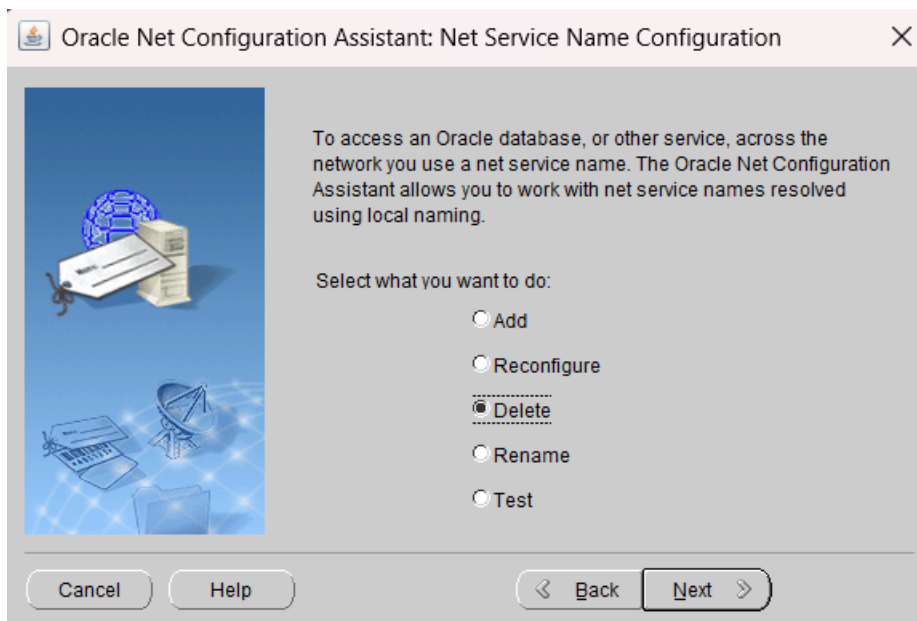


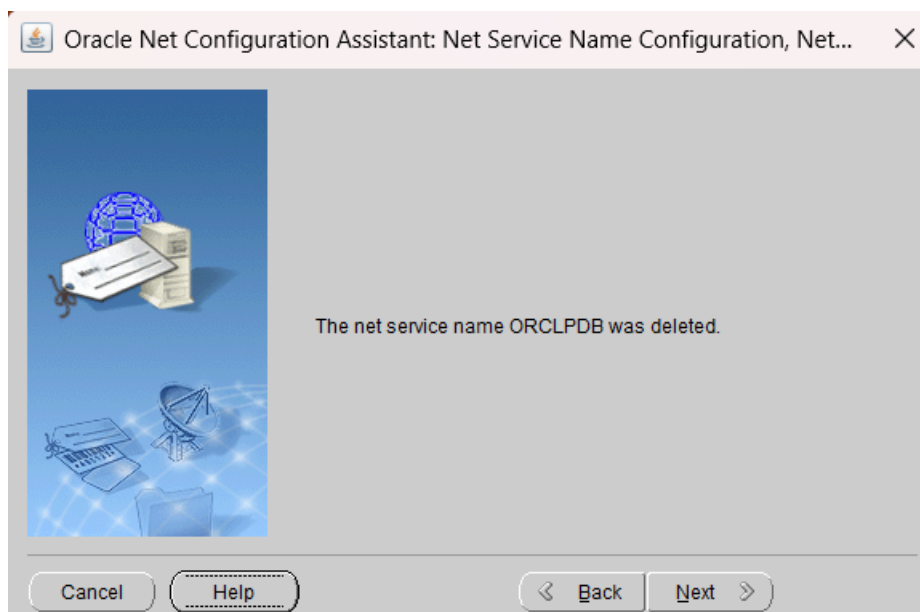
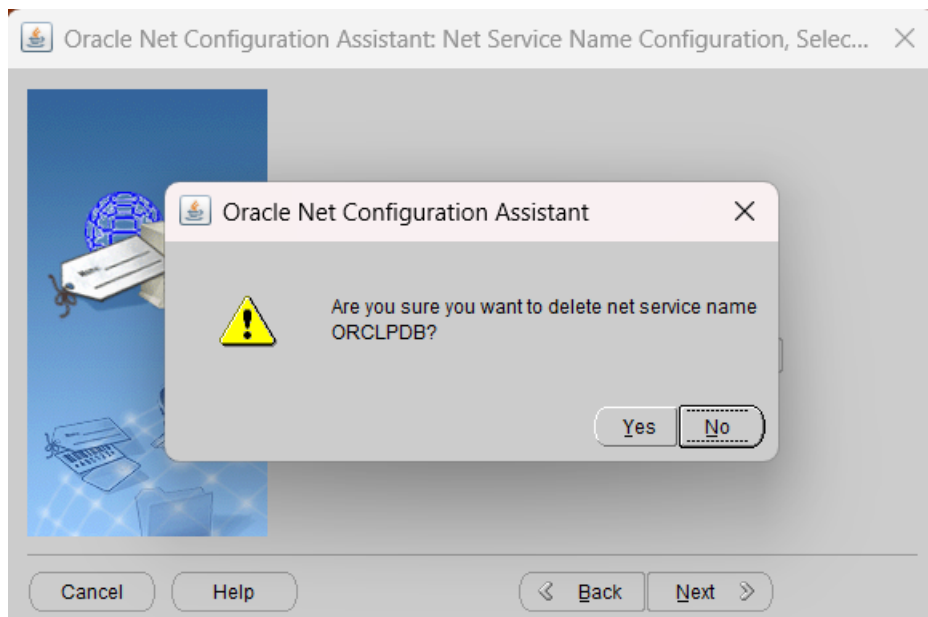
Step 5: Test the Local Net Service Name

```
SQL> connect system/thaxaena@orclpdb;  
Connected.  
SQL> show con_name;  
  
CON_NAME  
-----  
ORCLPDB
```

Step 6: Delete the Local Net Service Name







Verify delete

```
SQL> connect system/thaxaena@orclpdb;
ERROR:
ORA-12154: TNS:could not resolve the connect identifier specified

Warning: You are no longer connected to ORACLE.
```

Part C: Listener Control Commands

Step 7: Start the Listener

```
C:\Users\VICTUS\suresh>lsnrctl start

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:28:06

Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01106: Listener using listener name LISTENER has already been started
```

Step 8: Check Listener Status

```
C:\Users\VICTUS\suresh>lsnrctl status

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:28:28

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date                21-JUN-2026 08:45:07
Uptime                    21 days 6 hr. 43 min. 21 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   C:\Oracle_19C\network\admin\listener.ora
Listener Log File         C:\Oracle19C\diag\tnslsnr\Aconitin\listener\alert\log.xml
```

Step 9: Reload the Listener

```
C:\Users\VICTUS\suresh>lsnrctl reload

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:30:44

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
The command completed successfully
```

Step 10: Stop the Listener

```
C:\Users\VICTUS\suresh>lsnrctl stop

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:31:02

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
The command completed successfully
```

4. UNLOCKING HR-SCHEMA IN ORACLE 19C AND CONNECTING WITH HR USER.

THEORY:

HR Schema

The **HR (Human Resources) Schema** is Oracle's sample database schema used for learning and practicing SQL, PL/SQL, and database administration. It contains tables related to employees, departments, jobs, locations, countries, regions, and job history.

By default, the HR account is **locked** after Oracle installation for security reasons. It must be unlocked before it can be used

TNS (Transparent Network Substrate) Service

A **TNS Service** is a local network service name that allows Oracle clients to connect to a database using an easy-to-remember alias instead of specifying the complete connection details. It is configured using **Oracle Net Configuration Assistant (NetCA)** or manually through the tnsnames.ora file.

Example:

```
CONNECT hr/password@ORCLPDB;
```

Pluggable Database (PDB)

Oracle 19c uses **Multitenant Architecture**, where a **Container Database (CDB)** contains one or more **Pluggable Databases (PDBs)**. Since the HR schema resides inside a PDB, the session must be switched to the appropriate PDB before unlocking the HR account.

ALTER USER Statement

The ALTER USER statement is used to modify the properties of an existing user account, such as changing the password, locking or unlocking the account, and assigning roles or privileges.

Syntax:

```
ALTER USER username IDENTIFIED BY password ACCOUNT UNLOCK;
```

STEPS:

Step 1: Create a TNS Service

Using **Oracle Net Configuration Assistant (NetCA)**:

1. Open **NetCA**.
2. Select **Local Net Service Name Configuration**.
3. Click **Add**.

4. Enter the Service Name (e.g., ORCLPDB).
5. Select **TCP** as the protocol.
6. Enter the Host Name (e.g., localhost).
7. Enter the Listener Port (1521).
8. Test the connection.
9. Assign the Net Service Name.
10. Finish the configuration.

Step 3: Connect as SYSDBA

SQLPlus / AS SYSDBA;

Step 4: Verify the Current Container

SHOW CON_NAME;

Step 5: Switch to the ORCLPDB Container

ALTER SESSION SET CONTAINER = ORCLPDB;

Step 6: Verify the Current Container

SHOW CON_NAME;

Step 7: Unlock the HR Account

Replace your_password with your own password.

ALTER USER hr IDENTIFIED BY your_password ACCOUNT UNLOCK;

Step 8: Disconnect from SYS User

DISCONNECT;

Step 9: Connect Using the HR User

```
CONNECT hr/password@ORCLPDB;
```

Step 10: Verify the Connected User

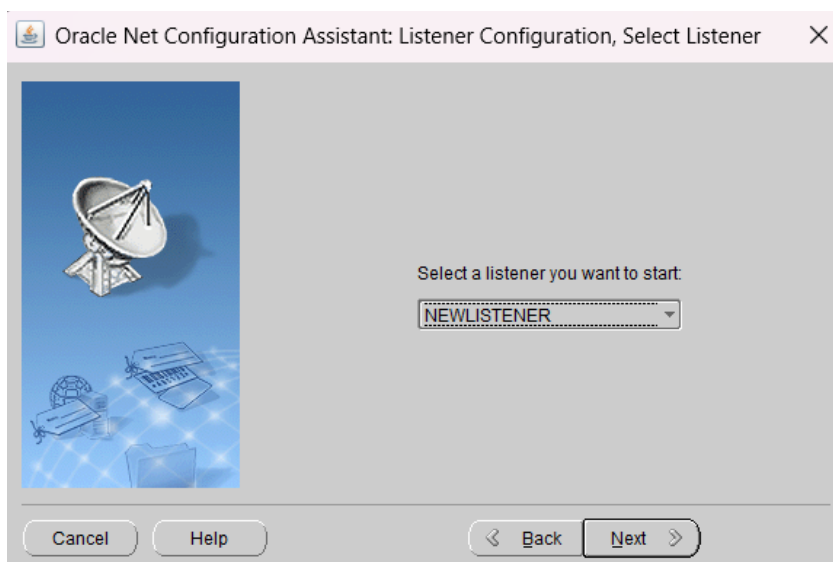
```
SHOW USER;
```

Step 11: Verify the Current Container

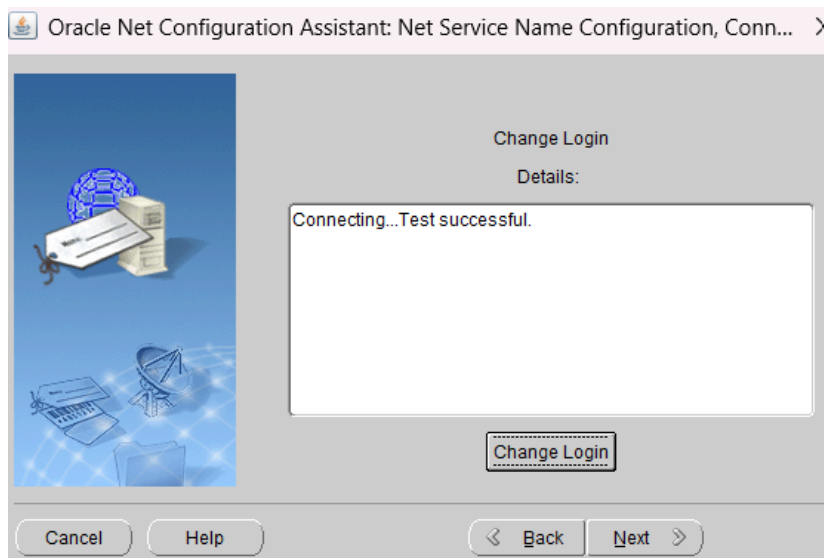
```
SHOW CON_NAME;
```

OUTPUT:

Step 1: Configure listener



Test the connection



Step 3: Connect as SYSDBA

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sun Jul 12 15:53:47 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 4: Verify the Current Container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

Step 5: Switch to the ORCLPDB Container and verify the connection name

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = orclpdb;

Session altered.

SQL> show con_name;

CON_NAME
-----
ORCLPDB
```

Step 7: Unlock the HR Account

```
SQL> alter user hr identified by thaxaena account unlock;

User altered.

SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

Step 8: Disconnect from SYS User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> disconnect
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 9: Connect Using the HR User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> disconnect
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
SQL> connect hr/thaxaena@orclpdb
Connected.
```

Step 10: Verify the Connected User

```
SQL> exec my_info;
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> show user;
USER is "HR"
```

Step 11: Verify the Current Container

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> show con_name

CON_NAME

ORCLPDB

5. DISPLAYING BASIC INFORMATION IN ORACLE DATABASE AND FORMATTING OUTPUT IN SQL PLUS.

THEORY:

Oracle provides various **data dictionary views** to retrieve information about database users, privileges, and tablespaces. These views help database administrators monitor and manage user accounts.

SQL*Plus also provides formatting commands that improve the readability of query results by adjusting line size, formatting column display, and limiting the number of rows displayed.

USER Information

Oracle stores information about user accounts such as username, account status, and default tablespace in the DBA_USERS data dictionary view.

Current User

The currently connected user can be displayed using the SHOW USER command or the USER function.

User Privileges

Privileges determine the actions a user can perform in the database. System privileges granted to users can be viewed using the DBA_SYS_PRIVS view.

Database Users

Information about all users present in the database can be viewed using the DBA_USERS view.

SQL*Plus Formatting

SQL*Plus provides formatting commands such as SET LINESIZE, COLUMN, and row limiting using FETCH FIRST or ROWNUM to improve the presentation of query results.

STEPS:

Steps and Commands

Step 1: View Username, Account Status, and Default Tablespace

```
SELECT USERNAME,  
       ACCOUNT_STATUS,  
       DEFAULT_TABLESPACE  
FROM DBA_USERS;
```

Step 2: Display the Current User

```
SHOW USER;
```

Step 3: Display Privileges Granted to a User

Replace HR with the desired username if needed.

```
SELECT GRANTEE,  
       PRIVILEGE  
FROM DBA_SYS_PRIVS  
WHERE GRANTEE = 'HR';
```

Step 4: Display All Users in the Database

```
SELECT USERNAME  
FROM DBA_USERS;
```

Formatting Output in SQL*Plus

Step 5: Set Line Size

```
SET LINESIZE 120;
```

Step 6: Format Column Display

```
COLUMN USERNAME FORMAT A20;  
COLUMN ACCOUNT_STATUS FORMAT A20;  
COLUMN DEFAULT_TABLESPACE FORMAT A20;
```

Step 7: Display Formatted Output

```
SELECT USERNAME,  
       ACCOUNT_STATUS,  
       DEFAULT_TABLESPACE  
FROM DBA_USERS;
```

Step 8: Limit the Number of Results

Using Oracle 12c/19c row limiting clause:

```
SELECT USERNAME,
       ACCOUNT_STATUS
FROM DBA_USERS
FETCH FIRST 5 ROWS ONLY;
```

or using ROWNUM:

```
SELECT USERNAME,
       ACCOUNT_STATUS
FROM DBA_USERS
WHERE ROWNUM <= 5;
```

OUTPUT:

Step 1: View Username, Account Status, and Default Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select username, account_status, default_tablespace from dba_users;
```

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
SYS	OPEN	SYSTEM
SYSTEM	OPEN	SYSTEM
XS\$NULL	EXPIRED & LOCKED	SYSTEM
OJVMSYS	LOCKED	SYSTEM
LBACSYS	LOCKED	SYSTEM
OUTLN	LOCKED	SYSTEM
SYS\$UMF	LOCKED	SYSTEM
DBSNMP	LOCKED	SYSAUX
APPQOSSYS	LOCKED	SYSAUX
DBSFUSER	LOCKED	SYSAUX
GGSYS	LOCKED	SYSAUX

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
ANONYMOUS	EXPIRED & LOCKED	SYSAUX
CTXSYS	EXPIRED & LOCKED	SYSAUX
DVSYS	LOCKED	SYSAUX

Step 2: Display the Current User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show user;
USER is "SYS"
```

Step 3: Display Privileges Granted to a User

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select grantee,privilege from dba_sys_privs where grantee='HR';

no rows selected
```

Step 4: Display All Users in the Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select username from dba_users;

USERNAME
-----
SYS
SYSTEM
XS$NULL
OJVMSYS
LBACSYS
OUTLN
SYS$UMF
DBSNMP
APPQOSSYS
DBSFUSER
GGSYS
ANONYMOUS
CTXSYS
DVSYS
DVF
GSMADMIN_INTERNAL
MDSYS
```

Formatting Output in SQL*Plus

Step 5: Set Line Size

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> set linesize 130;
```

Step 6: Format Column Display

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> set linesize 130;
SQL> COLUMN USERNAME FORMAT A20;
SQL> COLUMN ACCOUNT_STATUS FORMAT A20;
SQL> COLUMN DEFAULT_TABLESPACE FORMAT A20;
```

Step 7: Display Formatted Output

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT USERNAME,
2         ACCOUNT_STATUS,
3         DEFAULT_TABLESPACE
4 FROM DBA_USERS;
```

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
SYS	OPEN	SYSTEM
SYSTEM	OPEN	SYSTEM
XS\$NULL	EXPIRED & LOCKED	SYSTEM
OJVMSYS	LOCKED	SYSTEM
LBACSYS	LOCKED	SYSTEM
OUTLN	LOCKED	SYSTEM
SYS\$UMF	LOCKED	SYSTEM
DBSNMP	LOCKED	SYSAUX
APPQOSSYS	LOCKED	SYSAUX
DBSFUSER	LOCKED	SYSAUX
GGSYS	LOCKED	SYSAUX
ANONYMOUS	EXPIRED & LOCKED	SYSAUX
CTXSYS	EXPIRED & LOCKED	SYSAUX
DVSY	LOCKED	SYSAUX

Step 8: Limit the Number of Results

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT USERNAME,
2         ACCOUNT_STATUS
3 FROM DBA_USERS
4 WHERE ROWNUM <= 5;
```

USERNAME	ACCOUNT_STATUS
SYS	OPEN
AUDSYS	LOCKED
SYSTEM	OPEN
SYSBACKUP	LOCKED
SYSDG	LOCKED

6. WORKING WITH TABLESPACE AND DATA FILE

THEORY:

Tablespace

A tablespace is a logical storage unit in an Oracle database that stores database objects such as tables, indexes, and views. Each tablespace consists of one or more physical data files stored on disk.

Oracle mainly uses three types of tablespaces:

- Permanent Tablespace – Stores permanent database objects such as tables and indexes.
- Temporary Tablespace – Stores temporary data generated during sorting and query processing.
- Undo Tablespace – Stores undo information used for transaction rollback and recovery.

DBA_TABLESPACES View

DBA_TABLESPACES is a data dictionary view that displays information about all tablespaces in the database. It is accessible only to users with DBA privileges.

USER_TABLESPACES View

USER_TABLESPACES displays information about the tablespaces accessible to the currently connected user.

V\$TABLESPACE View

V\$TABLESPACE is a dynamic performance view that provides real-time information about the current state of tablespaces in the database.

DBA_DATA_FILES View

DBA_DATA_FILES displays detailed information about all data files belonging to permanent tablespaces, including their names, sizes, and associated tablespaces.

V\$DATAFILE View

V\$DATAFILE is a dynamic performance view that provides real-time information about data files currently being used by the Oracle database.

STEPS:

(a) Displaying Tablespaces

Step 1: Display All Tablespaces Using DBA_TABLESPACES

```
SELECT TABLESPACE_NAME,  
       STATUS,  
       CONTENTS  
FROM DBA_TABLESPACES;
```

Step 2: Login as HR User

```
CONNECT hr/password@ORCLPDB;
```

Step 3: Display Tablespace Associated with Current User

```
SELECT TABLESPACE_NAME,  
       STATUS  
FROM USER_TABLESPACES;
```

Step 4: Connect Back as SYSDBA

```
CONNECT / AS SYSDBA;
```

Step 5: Display Real-Time Information Using V\$TABLESPACE

```
SELECT TS#,  
       NAME  
FROM V$TABLESPACE;
```

Step 6: Display Tablespaces with Their Types

```
SELECT TABLESPACE_NAME,  
       CONTENTS  
FROM DBA_TABLESPACES;
```

(b) Create a Permanent Tablespace and Display the Newly Created Tablespace

Step 7: Connect as SYSDBA

CONNECT / AS SYSDBA;

Step 8: Verify the Current Container

SHOW CON_NAME;

If the current container is not ORCLPDB, switch to it:

ALTER SESSION SET CONTAINER = ORCLPDB;

Step 9: Create a Permanent Tablespace

Using your Oracle installation path:

```
CREATE TABLESPACE suresh_tbs1_23
DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'
SIZE 20M
AUTOEXTEND ON
NEXT 5M
MAXSIZE 100M;
```

Step 10: Display the Newly Created Tablespace

```
SELECT TABLESPACE_NAME,
       STATUS,
       CONTENTS
FROM DBA_TABLESPACES
WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

Step 11: Display the List of Tablespaces

```
SELECT TABLESPACE_NAME,
       CONTENTS
```

```
FROM DBA_TABLESPACES  
ORDER BY TABLESPACE_NAME;
```

(c) Modify the Tablespace Created Above

Step 12: Change the Tablespace Status to OFFLINE

```
ALTER TABLESPACE suresh_tbs1_23 OFFLINE;
```

Step 13: Verify the Tablespace Status

```
SELECT TABLESPACE_NAME,  
       STATUS  
FROM DBA_TABLESPACES  
WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

Step 14: Change the Tablespace Status to ONLINE

```
ALTER TABLESPACE suresh_tbs1_23 ONLINE;
```

Step 15: Verify the Tablespace Status Again

```
SELECT TABLESPACE_NAME,  
       STATUS  
FROM DBA_TABLESPACES  
WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

Step 1: 6Add a New Data File to the Tablespace

```
ALTER TABLESPACE suresh_tbs1_23  
ADD DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df2_23.dbf'  
SIZE 20M  
AUTOEXTEND ON  
NEXT 5M  
MAXSIZE 100M;
```

Step 17: Verify the Data Files

```
SELECT TABLESPACE_NAME,  
       FILE_NAME  
FROM DBA_DATA_FILES  
WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

Step 18: Rename the Tablespace

```
ALTER TABLESPACE suresh_tbs1_23  
RENAME TO suresh_dahal_tbs1_23;
```

Step 19: Verify the Renamed Tablespace

```
SELECT TABLESPACE_NAME  
FROM DBA_TABLESPACES  
WHERE TABLESPACE_NAME = 'SURESH_DAHAL_TBS1_23';
```

(d) Create and Delete a Tablespace**Step 20: Create a New Tablespace**

```
CREATE TABLESPACE test_tbs_by_suresh  
DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\test_df_by_suresh.dbf'  
SIZE 20M  
AUTOEXTEND ON  
NEXT 5M  
MAXSIZE 100M;
```

Step 21: Display the Newly Created Tablespace

```
SELECT TABLESPACE_NAME,  
       STATUS,  
       CONTENTS  
FROM DBA_TABLESPACES  
WHERE TABLESPACE_NAME = 'TEST_TBS_BY_SURESH';
```

Step 22: Display the List of Tablespaces

```
SELECT TABLESPACE_NAME  
FROM DBA_TABLESPACES  
ORDER BY TABLESPACE_NAME;
```

Step 23: Delete the Tablespace

```
DROP TABLESPACE test_tbs_by_suresh  
INCLUDING CONTENTS  
AND DATAFILES;
```

Step 24: Display the List of Tablespaces Again

```
SELECT TABLESPACE_NAME  
FROM DBA_TABLESPACES  
ORDER BY TABLESPACE_NAME;
```

e) Displaying Data Files**Step 25: Display Information About Data Files Using DBA_DATA_FILES**

```
SELECT FILE_NAME,
```

```
TABLESPACE_NAME,  
BYTES,  
AUTOEXTENSIBLE,  
STATUS  
FROM DBA_DATA_FILES;
```

This displays information about all data files in the database.

Step 26: Display Information About Data Files Using V\$DATAFILE

```
SELECT FILE#,  
NAME,  
STATUS  
FROM V$DATAFILE;
```

This displays real-time information about the data files currently being used by the database.

f) Resize the Data File Created in (b)

Step 27: Resize the Data File

```
ALTER DATABASE  
DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'  
RESIZE 50M;
```

Step 28: Verify the Resized Data File

```
SELECT FILE_NAME,  
BYTES/1024/1024 AS SIZE_MB  
FROM DBA_DATA_FILES  
WHERE FILE_NAME LIKE '%SURESH_DF1_23.DBF';
```

OUTPUT:

(a) Displaying Tablespaces

Step 1: Display All Tablespaces Using DBA_TABLESPACES

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.
```

```
SQL> SELECT TABLESPACE_NAME,
2          STATUS,
3          CONTENTS
4  FROM DBA_TABLESPACES;
```

TABLESPACE_NAME	STATUS	CONTENTS
SYSTEM	ONLINE	PERMANENT
SYSAUX	ONLINE	PERMANENT
UNDOTBS1	ONLINE	UNDO
TEMP	ONLINE	TEMPORARY
USERS	ONLINE	PERMANENT

Step 2: Login as HR User

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> connect hr/thaxaena@orclpdb
Connected.
```

Step 3: Display Tablespace Associated with Current User

```
SQL> exec my_info
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.
```

```
SQL> SELECT TABLESPACE_NAME,
2          STATUS
3  FROM USER_TABLESPACES;
```

TABLESPACE_NAME	STATUS
SYSTEM	ONLINE
SYSAUX	ONLINE
UNDOTBS1	ONLINE
TEMP	ONLINE
USERS	ONLINE

Step 4: Connect Back as SYSDBA

```
SQL> exec my_info
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> connect / as sysdba
Connected.
```

Step 5: Display Real-Time Information Using V\$TABLESPACE

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TS#,
2         NAME
3 FROM V$TABLESPACE;

TS# NAME
-----
1 SYSAUX
0 SYSTEM
2 UNDOTBS1
4 USERS
3 TEMP
0 SYSTEM
1 SYSAUX
2 UNDOTBS1
3 TEMP
0 SYSTEM
1 SYSAUX
2 UNDOTBS1
3 TEMP
5 USERS
```

Step 6: Display Tablespaces with Their Types

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         CONTENTS
3 FROM DBA_TABLESPACES;

TABLESPACE_NAME      CONTENTS
-----
SYSTEM              PERMANENT
SYSAUX              PERMANENT
UNDOTBS1            UNDO
TEMP                TEMPORARY
USERS              PERMANENT
```

(b) Create a Permanent Tablespace and Display the Newly Created Tablespace

Step 7: Connect as SYSDBA

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> connect / as sysdba;
Connected.
```

Step 8: Verify the Current Container

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name

CON_NAME
-----
CDB$ROOT
SQL> alter session set container = orclpdb;

Session altered.
```

Step 9: Create a Permanent Tablespace

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> CREATE TABLESPACE suresh_tbs1_23
 2 DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'
 3 SIZE 20M
 4 AUTOEXTEND ON
 5 NEXT 5M
 6 MAXSIZE 100M;

Tablespace created.
```

Step 10: Display the Newly Created Tablespace

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
 2 STATUS,
 3 CONTENTS
 4 FROM DBA_TABLESPACES
 5 WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          STATUS          CONTENTS
-----
SURESH_TBS1_23           ONLINE         PERMANENT
```

Step 11: Display the List of Tablespaces

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
 2 CONTENTS
 3 FROM DBA_TABLESPACES
 4 ORDER BY TABLESPACE_NAME;

TABLESPACE_NAME          CONTENTS
-----
SURESH_TBS1_23           PERMANENT
SYSAUX                   PERMANENT
SYSTEM                   PERMANENT
TEMP                     TEMPORARY
UNDOTBS1                 UNDO
USERS                     PERMANENT
```

(c) Modify the Tablespace Created Above

Step 12: Change the Tablespace Status to OFFLINE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter tablespace suresh_tbs1_23 offline;

Tablespace altered.
```

Step 13: Verify the Tablespace Status

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         STATUS
3   FROM DBA_TABLESPACES
4  WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

TABLESPACE_NAME	STATUS
SURESH_TBS1_23	OFFLINE

Step 14: Change the Tablespace Status to ONLINE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23 ONLINE;

Tablespace altered.
```

Step 15: Verify the Tablespace Status Again

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         STATUS
3   FROM DBA_TABLESPACES
4  WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';
```

TABLESPACE_NAME	STATUS
SURESH_TBS1_23	ONLINE

Step 16: Add a New Data File to the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23
2  ADD DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df2_23.dbf'
3  SIZE 20M
4  AUTOEXTEND ON
5  NEXT 5M
6  MAXSIZE 100M;

Tablespace altered.
```

Step 17: Verify the Data Files

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         FILE_NAME
3 FROM DBA_DATA_FILES
4 WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          FILE_NAME
-----
SURESH_TBS1_23           C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB
SURESH_TBS1_23           C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DB
```

Step 18: Rename the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23
2  RENAME TO suresh_dahal_tbs1_23;

Tablespace altered.
```

Step 19: Verify the Renamed Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
2 FROM DBA_TABLESPACES
3 WHERE TABLESPACE_NAME = 'SURESH_DAHAL_TBS1_23';

TABLESPACE_NAME
-----
SURESH_DAHAL_TBS1_23
```

(d) Create and Delete a Tablespace

Step 20: Create a New Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> CREATE TABLESPACE test_tbs_by_suresh
2 DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\test_df_by_suresh.dbf'
3 SIZE 20M
4 AUTOEXTEND ON
5 NEXT 5M
6 MAXSIZE 100M;

Tablespace created.
```

Step 21: Display the Newly Created Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         STATUS,
3         CONTENTS
4 FROM DBA_TABLESPACES
5 WHERE TABLESPACE_NAME = 'TEST_TBS_BY_SURESH';
```

TABLESPACE_NAME	STATUS	CONTENTS
TEST_TBS_BY_SURESH	ONLINE	PERMANENT

Step 22: Display the List of Tablespaces

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
2 FROM DBA_TABLESPACES
3 ORDER BY TABLESPACE_NAME;
```

TABLESPACE_NAME
SURESH_DAHAL_TBS1_23
SYSAUX
SYSTEM
TEMP
TEST_TBS_BY_SURESH
UNDOTBS1
USERS

Step 23: Delete the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> DROP TABLESPACE test_tbs_by_suresh
2 INCLUDING CONTENTS
3 AND DATAFILES;
```

Tablespace dropped.

Step 24: Display the List of Tablespaces Again

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
       2 FROM DBA_TABLESPACES
       3 ORDER BY TABLESPACE_NAME;

TABLESPACE_NAME
-----
SURESH_DAHAL_TBS1_23
SYSAUX
SYSTEM
TEMP
UNDOTBS1
USERS
```

e) Displaying Data Files

Step 25: Display Information About Data Files Using DBA_DATA_FILES

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT FILE_NAME,
       2     TABLESPACE_NAME,
       3     BYTES,
       4     AUTOEXTENSIBLE,
       5     STATUS
       6 FROM DBA_DATA_FILES;

FILE_NAME                                TABLESPACE_NAME      BYTES  AUT  STATUS
-----
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF  SYSTEM                283115520 YES  AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSAUX01.DBF  SYSAUX                335544320 YES  AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF  UNDOTBS1              104857600 YES  AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF    USERS                  5242880 YES  AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB SURESH_DAHAL_TBS1_23  20971520 YES  AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DB SURESH_DAHAL_TBS1_23  20971520 YES  AVAILABLE

6 rows selected.
```

Step 26: Display Information About Data Files Using V\$DATAFILE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT FILE#,
       2     NAME,
       3     STATUS
       4 FROM V$DATAFILE;

FILE# NAME                                STATUS
-----
9 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF  SYSTEM
10 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSAUX01.DBF  ONLINE
11 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF  ONLINE
12 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF    ONLINE
13 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DBF  ONLINE
14 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DBF  ONLINE

6 rows selected.
```

f) Resize the Data File Created in (b)

Step 27: Resize the Data File

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER DATABASE
  2  DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'
  3  RESIZE 50M;

Database altered.
```

Step 28: Verify the Resized Data File

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT FILE_NAME,
  2         BYTES/1024/1024 AS SIZE_MB
  3  FROM DBA_DATA_FILES
  4  WHERE FILE_NAME LIKE '%SURESH_DF1_23.DBF';
```

FILE_NAME	SIZE_MB
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB	50

7. WORKING WITH ORACLE SQL DEVELOPER

THEORY:

Oracle SQL Developer

Oracle SQL Developer is a free graphical database development tool provided by Oracle for working with Oracle databases. It enables users to create database connections, execute SQL and PL/SQL statements, manage database objects, browse schemas, and perform database administration tasks through an easy-to-use graphical interface.

Some features of Oracle SQL Developer include:

- Creating and managing database connections.
- Executing SQL and PL/SQL statements using SQL Worksheet.
- Browsing database objects such as tables, views, indexes, and procedures.
- Importing and exporting database data.
- Managing users and database administration tasks.
- Generating reports and viewing query results.

STEPS:

Step 1: Install Oracle SQL Developer

1. Download Oracle SQL Developer.
2. Install the software.
3. Launch Oracle SQL Developer.

Part A: Create a Connection for SYSTEM User

Step 2: Create a New Database Connection

1. Click **New Connection (+)**.
2. Enter the following details:

Field	Value
Connection Name	SYSTEM
Username	system

Field	Value
Password	Pass123
Connection Type	Basic
Hostname	localhost
Port	1521
Service Name	ORCLPDB

3. Click **Test**.
4. Ensure the status shows **Success**.
5. Click **Connect**.

Step 3: Execute Commands from SQL Worksheet

SHOW USER;

SHOW CON_NAME;

SELECT TABLESPACE_NAME

FROM DBA_TABLESPACES;

Part B: Create a Connection for HR User

Step 4: Create a New Database Connection

1. Click **New Connection (+)**.
2. Enter the following details:

Field	Value
Connection Name	HR
Username	hr
Password	Your HR Password
Connection Type	Basic

Field	Value
Hostname	localhost
Port	1521
Service Name	ORCLPDB (<i>or your PDB name</i>)

3. Click **Test**.
4. Ensure the status shows **Success**.
5. Click **Connect**.

Step 5: Execute Commands from SQL Worksheet

```
SHOW USER;
SHOW CON_NAME;
SELECT EMPLOYEE_ID,
       FIRST_NAME,
       LAST_NAME
FROM EMPLOYEES;
```

Step 6: Execute Any Additional Query

```
SELECT DEPARTMENT_ID,
       DEPARTMENT_NAME
FROM DEPARTMENTS;
```

OUTPUT:

Step 1: Install Oracle SQL Developer



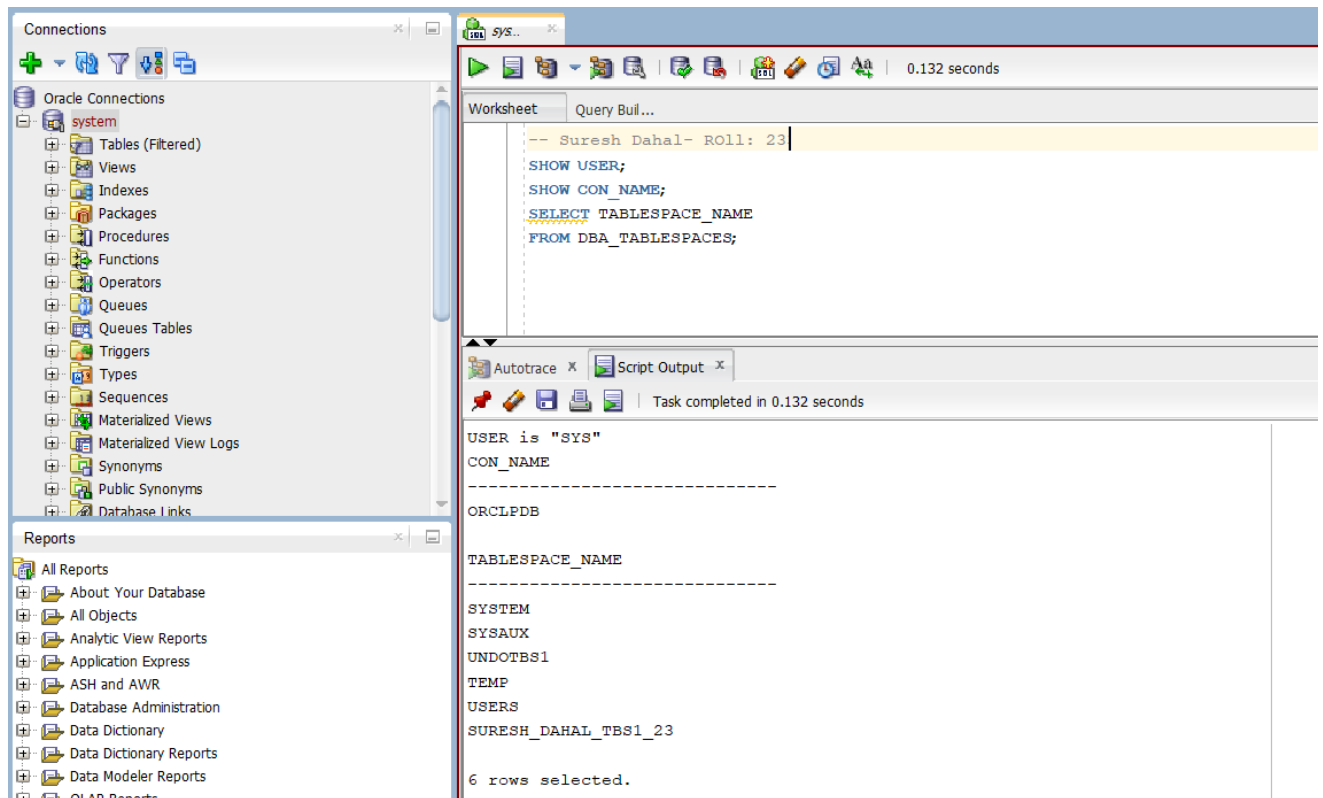
Part A: Create a Connection for SYSTEM User

Step 2: Create a New Database Connection

New / Select Database Connection

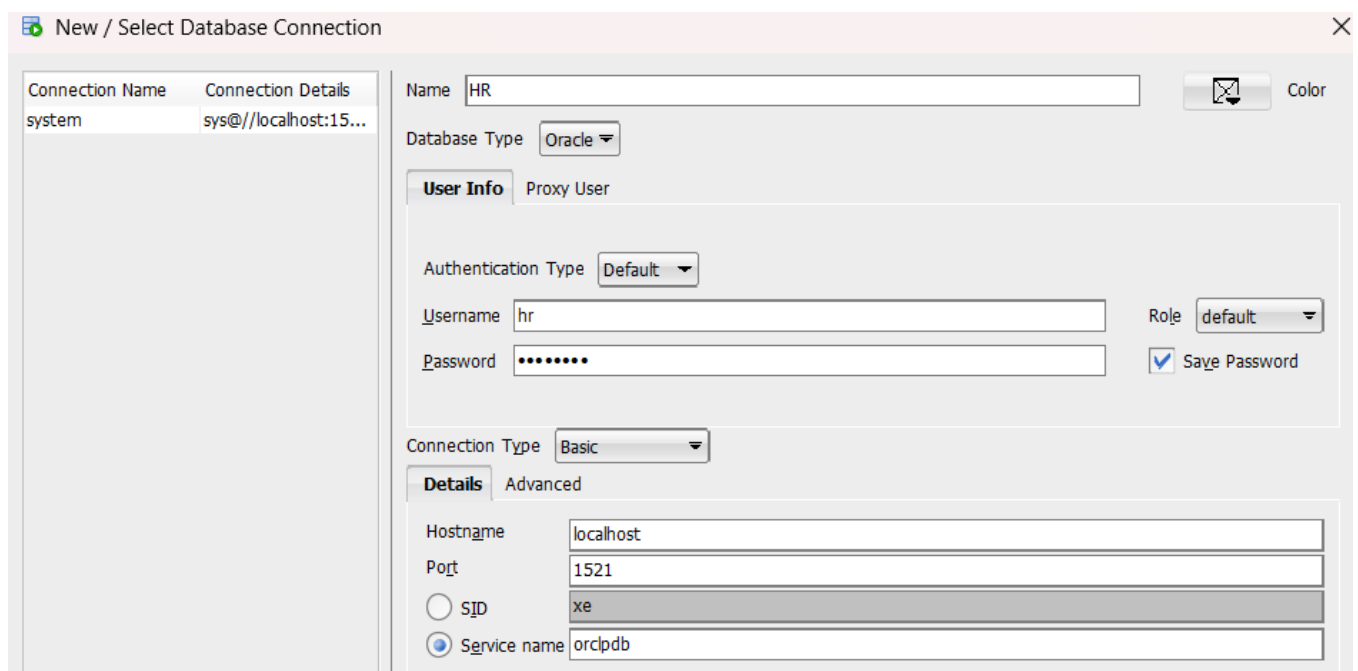
Connection Name	Connection Details
	Name: system Database Type: Oracle User Info Proxy User Authentication Type: Default Username: sys Role: SYSDBA Password: ☒ Save Password Connection Type: Basic Details Advanced Hostname: localhost Port: 1521 ☐ SID xe ☒ Service name orclpdb

Step 3: Execute Commands from SQL Worksheet



Part B: Create a Connection for HR User

Step 4: Create a New Database Connection



Step 5: Execute Commands from SQL Worksheet

The screenshot displays the SQL Developer interface with the following components:

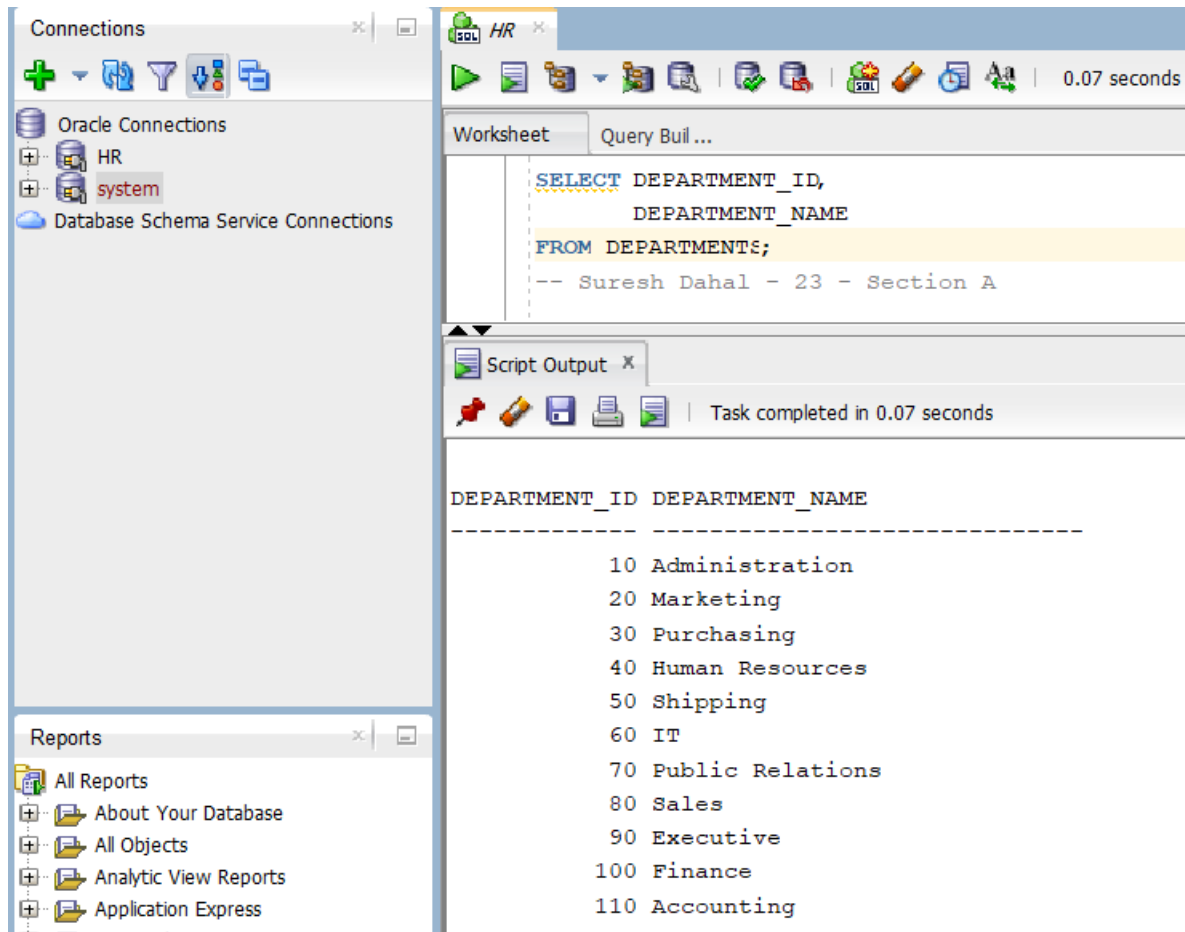
- Connections:** Shows Oracle Connections with 'HR' and 'system' selected, and Database Schema Service Connections.
- Worksheet:** Contains the following SQL commands:

```
SHOW USER;
SHOW CON_NAME;
SELECT EMPLOYEE_ID,
       FIRST_NAME,
       LAST_NAME
FROM EMPLOYEES;
-- Suresh Dahal - 23 - Section A
```
- Script Output:** Displays the results of the executed commands:

```
USER is "HR"
CON_NAME
-----
ORCLPDB
```
- Reports:** A sidebar menu with various report categories.
- Results:** A table showing the output of the SELECT statement:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME
100	Steven	King
101	Neena	Kochhar
102	Lex	De Haan
103	Alexander	Hunold
104	Bruce	Ernst
105	David	Austin
106	Valli	Pataballa
107	Diana	Lorentz
108	Nancy	Greenberg
109	Daniel	Faviet
110	John	Chen

Step 6: Execute Any Additional Query



The screenshot displays the Oracle SQL Developer interface. On the left, the 'Connections' pane shows 'Oracle Connections' with 'HR' and 'system' connections. The 'Reports' pane at the bottom left shows 'All Reports' and 'Application Express'. The main 'Worksheet' pane shows a SQL query: `SELECT DEPARTMENT_ID, DEPARTMENT_NAME FROM DEPARTMENTS;` with a comment `-- Suresh Dahal - 23 - Section A`. The 'Script Output' pane shows the execution result: 'Task completed in 0.07 seconds' and a table of department data.

DEPARTMENT_ID	DEPARTMENT_NAME
10	Administration
20	Marketing
30	Purchasing
40	Human Resources
50	Shipping
60	IT
70	Public Relations
80	Sales
90	Executive
100	Finance
110	Accounting

8. USER MANAGEMENT WITH GRANT AND REVOKE PRIVILEGES.

THEORY:

User Management in Oracle

User management in Oracle is the process of creating, modifying, and managing database user accounts and their privileges. Oracle provides SQL statements such as CREATE USER, ALTER USER, DROP USER, GRANT, and REVOKE to perform user administration. Proper user management ensures database security by allowing users to access only the resources and operations for which they have been granted permissions.

DBA_USERS View

DBA_USERS is a data dictionary view that contains information about all users in the Oracle database. It provides details such as the username, account status, default tablespace, temporary tablespace, profile, and account creation date. This view is accessible only to users with DBA privileges.

STEPS:

(a) Display Information About Users Using DBA_USERS

Step 1: Display Information About Users

```
SELECT USERNAME,  
       ACCOUNT_STATUS,  
       DEFAULT_TABLESPACE  
FROM DBA_USERS;
```

b) Create a User

Step 4: Create User user_suresh

```
CREATE USER user_suresh  
IDENTIFIED BY pass123;
```

(c) Grant Privileges to the User

Step 5: Grant System Privileges

```
GRANT CREATE SESSION,  
       CREATE TABLE  
TO user_suresh;
```

(d) Create a Table After Logging in as user_suresh

Step 6: Connect as user_suresh

```
CONNECT user_suresh/pass123@ORCLPDB;
```

Step 7: Create a Table

```
CREATE TABLE student (  
    id NUMBER PRIMARY KEY,  
    name VARCHAR2(50),  
    address VARCHAR2(100)  
);
```

(e) Create Another User with Tablespace Clause**Step 8: Create User suresh_dahal**

```
CREATE USER suresh_dahal  
IDENTIFIED BY pass123  
DEFAULT TABLESPACE USERS  
TEMPORARY TABLESPACE TEMP;
```

(f) Create Another User of Your Choice**Step 10: Create User**

```
CREATE USER test_user  
IDENTIFIED BY pass123;
```

(g) Grant Privileges to the User Created in (f)**Step 11: Grant Privileges**

```
GRANT CREATE SESSION,  
    CREATE TABLE  
TO test_user;
```

(h) Change the Password of user_suresh**Step 12: Change Password**

```
ALTER USER user_suresh  
IDENTIFIED BY pass345;
```

(i) Enable Password Expiration for suresh_dahal**Step 13: Expire Password**

```
ALTER USER suresh_dahal
```

```
PASSWORD EXPIRE;
```

(j) Grant Object Privileges on the Table

Step 14: Connect as user_suresh

```
CONNECT user_suresh/pass345@ORCLPDB;
```

Step 15: Grant Object Privileges

```
GRANT SELECT,
```

```
INSERT,
```

```
UPDATE,
```

```
DELETE
```

```
ON student
```

```
TO suresh_dahal;
```

(k) Insert, Update and Select Records by Logging in as suresh_dahal

Step 16: Connect as suresh_dahal

```
CONNECT suresh_dahal/pass123@ORCLPDB;
```

Step 17: Insert Two Records

```
INSERT INTO user_suresh.student
```

```
VALUES (1, 'Ram', 'Kathmandu');
```

```
INSERT INTO user_suresh.student
```

```
VALUES (2, 'Hari', 'Pokhara');
```

Step 18: Update the Second Record

```
UPDATE user_suresh.student
```

```
SET address = 'Lalitpur'
```

```
WHERE id = 2;
```

Step 19: Display All Records

```
SELECT *
```

```
FROM user_suresh.student;
```

(l) Revoke DELETE Privilege

Step 20: Connect as user_suresh

```
CONNECT user_suresh/pass345@ORCLPDB;
```

Step 21: Revoke DELETE Privilege

```
REVOKE DELETE
```

```
ON student
```

```
FROM suresh_dahal;
```

(m) Remove the User Created in (f)

Step 22: Connect as SYSDBA

```
CONNECT / AS SYSDBA;
```

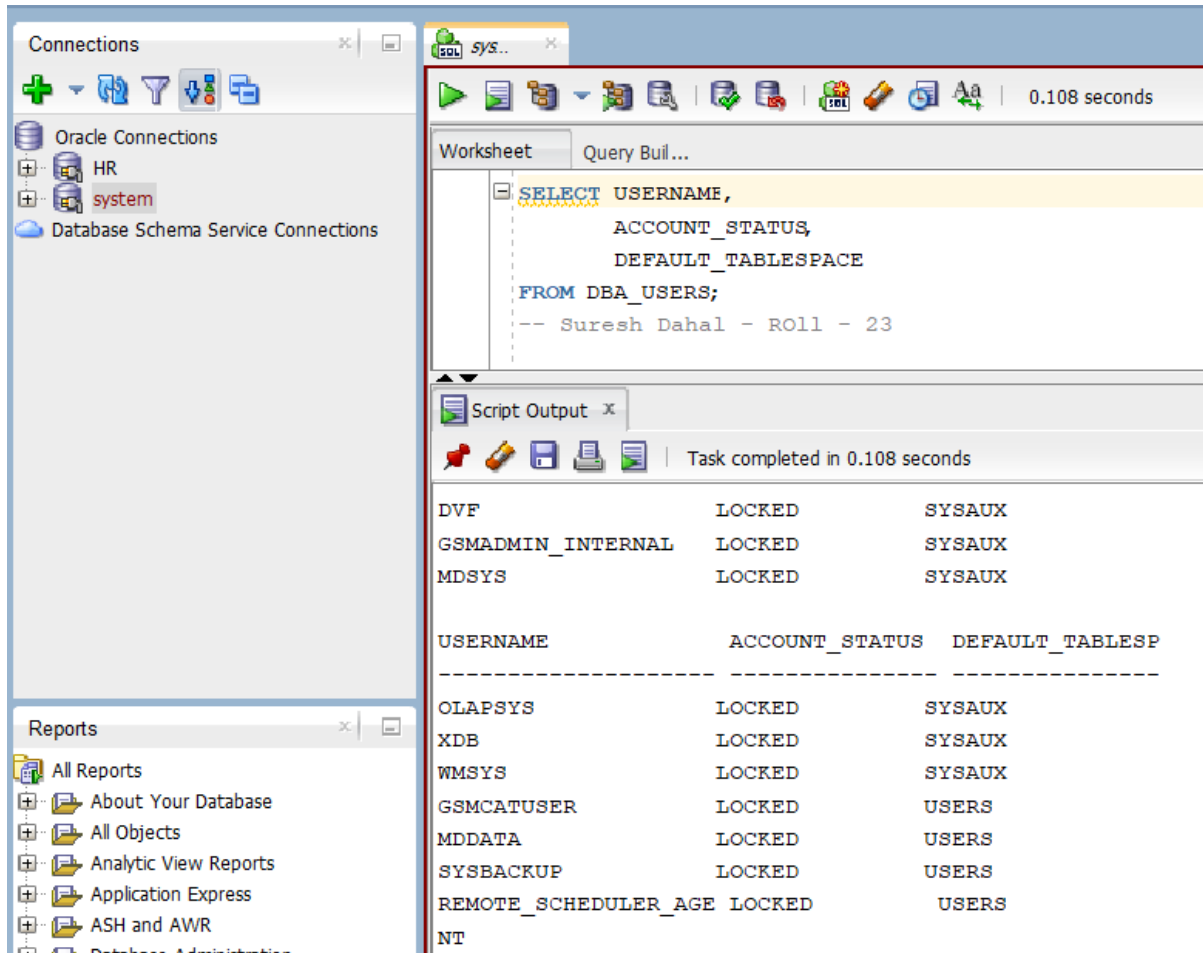
Step 23: Drop the User

```
DROP USER test_user CASCADE;
```

OUTPUT:

(a) Display Information About Users Using DBA_USERS

Step 1: Display Information About Users



The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane shows the 'system' connection selected. The 'Reports' pane is also visible. The main window displays a query in the 'Query Builder' tab:

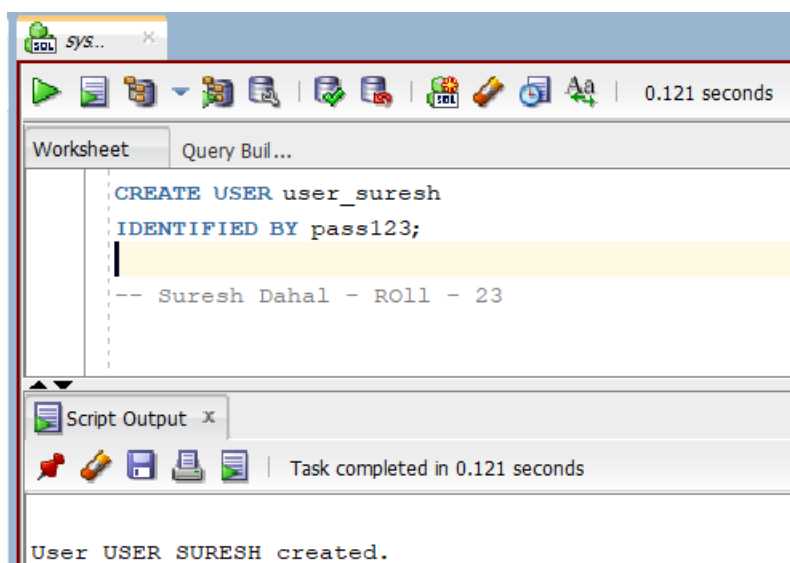
```
SELECT USERNAME,  
       ACCOUNT_STATUS,  
       DEFAULT_TABLESPACE  
FROM DBA_USERS;  
-- Suresh Dahal - Roll - 23
```

The 'Script Output' pane shows the results of the query, indicating it was completed in 0.108 seconds. The output is a table with three columns: USERNAME, ACCOUNT_STATUS, and DEFAULT_TABLESPACE.

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
DVF	LOCKED	SYSAUX
GSMADMIN_INTERNAL	LOCKED	SYSAUX
MDSYS	LOCKED	SYSAUX
OLAPSYS	LOCKED	SYSAUX
XDB	LOCKED	SYSAUX
WMSYS	LOCKED	SYSAUX
GSMCATUSER	LOCKED	USERS
MDDATA	LOCKED	USERS
SYSDATABASE	LOCKED	USERS
REMOTE_SCHEDULER_AGENT	LOCKED	USERS
NT		

b) Create a User

Step 4: Create User user_suresh



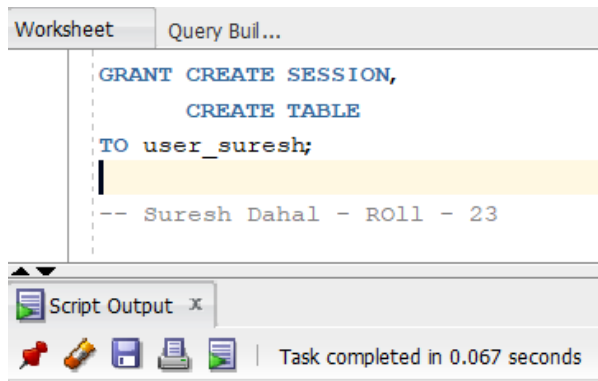
The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' tab displays the following SQL command:

```
CREATE USER user_suresh  
IDENTIFIED BY pass123;  
-- Suresh Dahal - Roll - 23
```

The 'Script Output' pane shows the results of the command, indicating it was completed in 0.121 seconds. The output is a single line: 'User USER_SURESH created.'

(c) Grant Privileges to the User

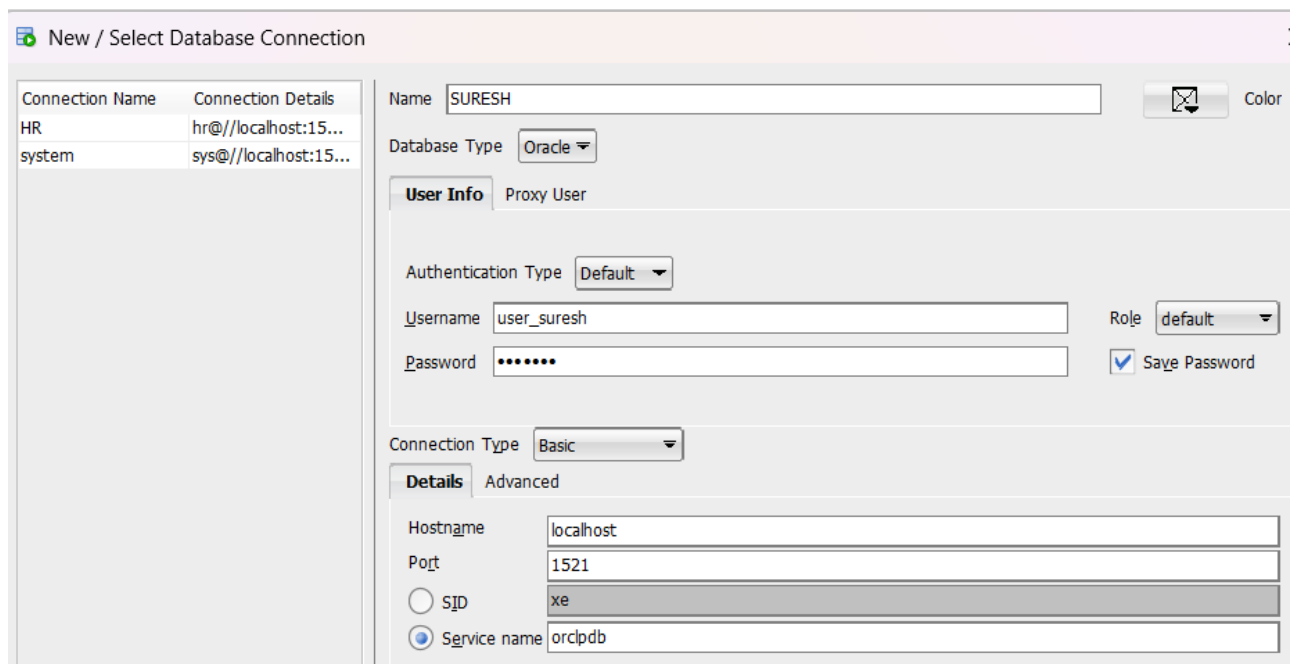
Step 5: Grant System Privileges



Grant succeeded.

(d) Create a Table After Logging in as user_suresh

Step 6: Connect as user_suresh



Step 7: Create a Table

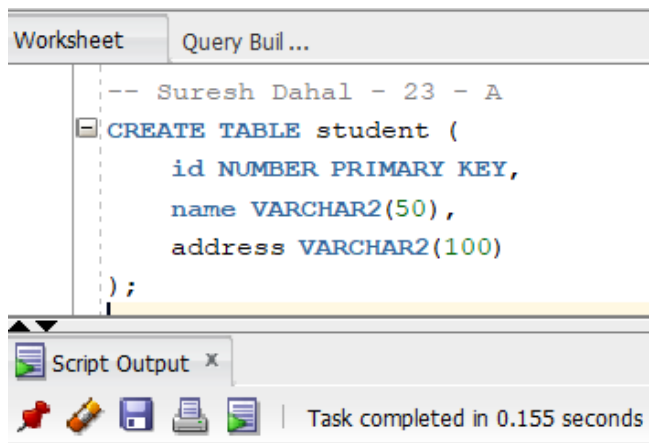
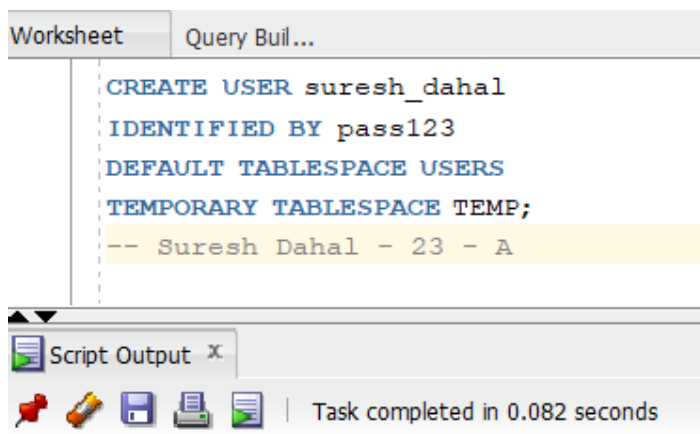


Table STUDENT created.

(e) Create Another User with Tablespace Clause

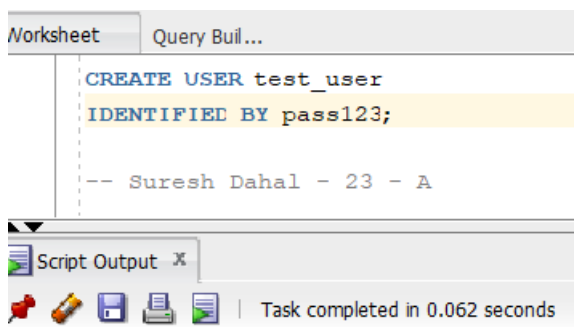
Step 8: Create User suresh_dahal



User SURESH_DAHAL created.

(f) Create Another User of Your Choice

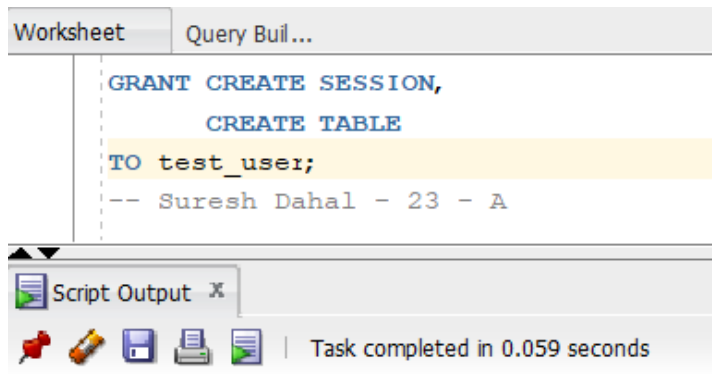
Step 10: Create User



User TEST_USER created.

(g) Grant Privileges to the User Created in (f)

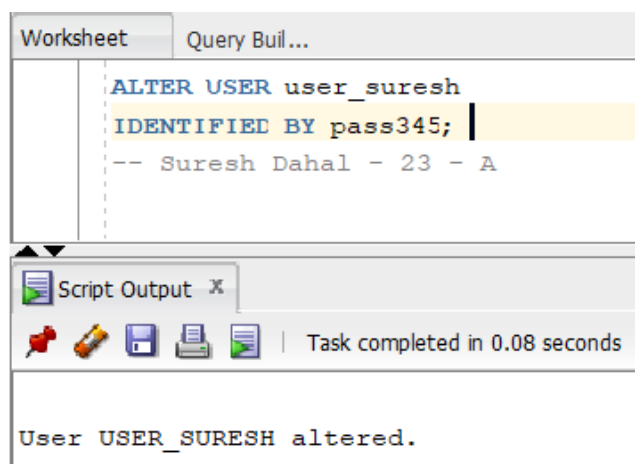
Step 11: Grant Privileges



Grant succeeded.

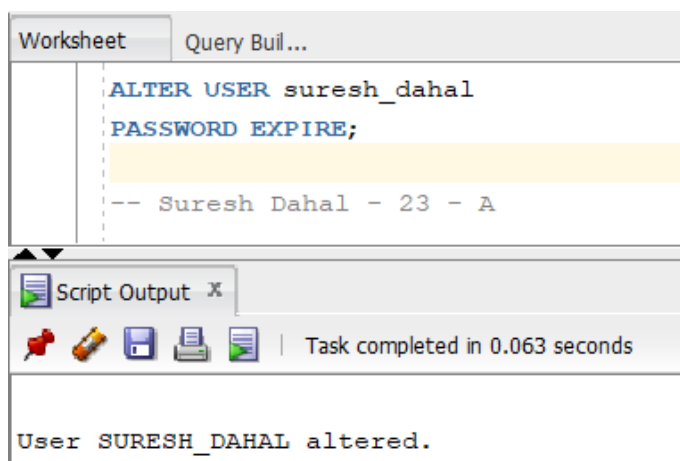
(h) Change the Password of user_suresh

Step 12: Change Password



(i) Enable Password Expiration for suresh_dahal

Step 13: Expire Password



(j) Grant Object Privileges on the Table

Step 14: Connect as user_suresh

New / Select Database Connection

Connection Name	Connection Details
HR	hr@//localhost:15...
SURESH	user_suresh@//loc...
system	sys@//localhost:15...

Name: User_Suresh

Database Type: Oracle

User Info: Proxy User

Authentication Type: Default

Username: user_suresh

Password:

Role: default

Save Password: ☐

Connection Type: Basic

Details: Advanced

Hostname: localhost

Port: 1521

☐ SID: xe

☒ Service name: orclpdb

Step 15: Grant Object Privileges

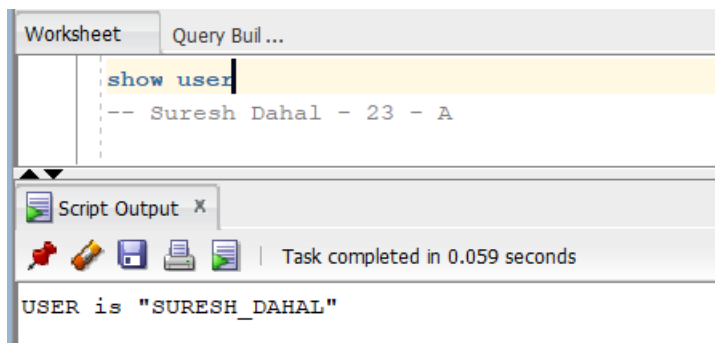
```
GRANT SELECT,
      INSERT,
      UPDATE,
      DELETE
ON student
TO suresh_dahal;
-- Suresh Dahal - 23 - A
```

Script Output: Task completed in 0.065 seconds

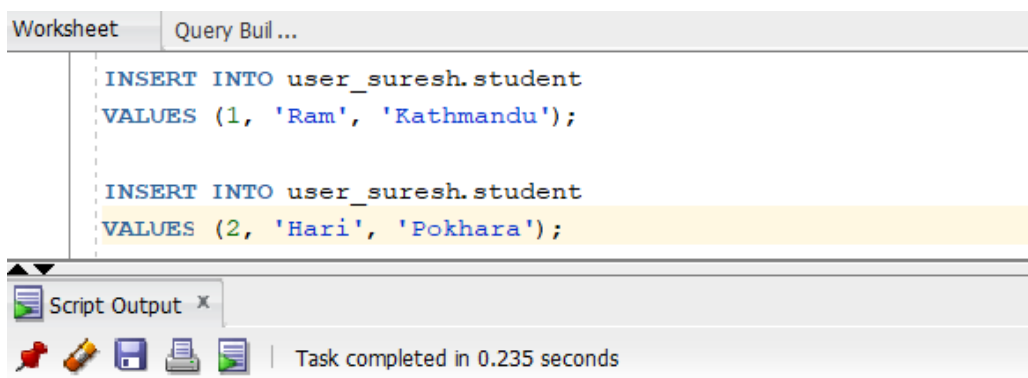
Grant succeeded.

(k) Insert, Update and Select Records by Logging in as suresh_dahal

Step 16: Connect as suresh_dahal



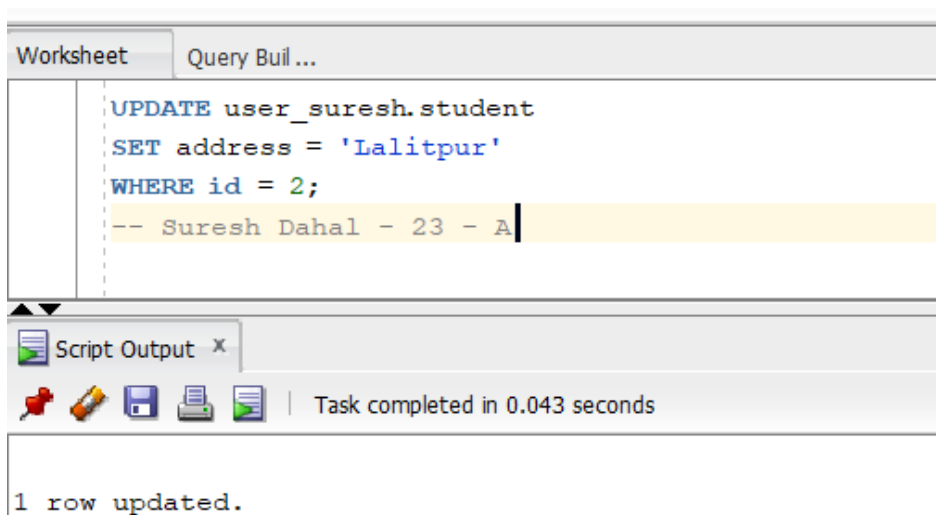
Step 17: Insert Two Records



1 row inserted.

1 row inserted.

Step 18: Update the Second Record



Step 19: Display All Records

Worksheet Query Buil...

```
SELECT *
FROM user_suresh.student;
-- Suresh Dahal - 23 - A
```

Script Output x

Task completed in 0.071 seconds

ID	NAME	ADDRESS
1	Ram	Kathmandu
2	Hari	Lalitpur

(l) Revoke DELETE Privilege

Step 20: Connect as user_suresh

Worksheet Query Buil...

```
show user
```

Script Output x

Task completed in 0.061 seconds

USER is "USER_SURESH"

Step 21: Revoke DELETE Privilege

Worksheet Query Buil...

```
REVOKE DELETE
ON student
FROM suresh_dahal;
-- Suresh Dahal - 23 - A
```

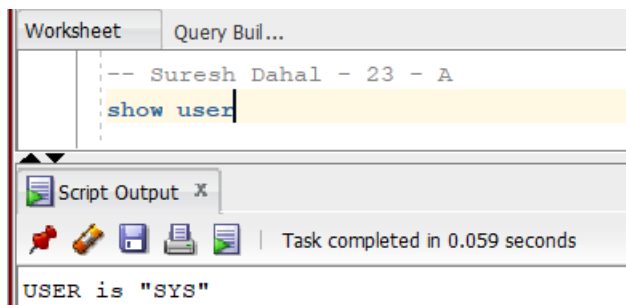
Script Output x

Task completed in 0.051 seconds

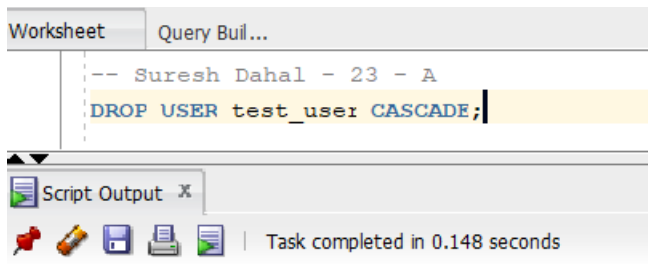
Revoke succeeded.

(m) Remove the User Created in (f)

Step 22: Connect as SYSDBA



Step 23: Drop the User



User TEST_USER dropped.

9. WORKING WITH ROLES AND PRIVILEGES.

THEORY:

A **role** is a collection of privileges that can be granted to one or more users. Roles simplify privilege management by allowing administrators to assign or revoke multiple privileges at once. Oracle supports **system privileges** such as CREATE TABLE and **object privileges** such as SELECT, INSERT, and UPDATE on specific database objects. Roles are managed using SQL statements like CREATE ROLE, GRANT, REVOKE, and DROP ROLE.

STEPS:

(a) Create a Role Named role_by_yourName

Step 1: Create the Role

```
CREATE ROLE role_by_suresh;
```

(b) Grant System and Object Privileges to the Role

Step 2: Grant a System Privilege

```
GRANT CREATE TABLE TO role_by_suresh;
```

Step 3: Connect as the Owner of the Table

```
CONNECT user_suresh/password@ORCLPDB;
```

Step 5: Grant Object Privileges

```
GRANT SELECT,
```

```
INSERT,
```

```
UPDATE
```

```
ON student
```

```
TO role_by_suresh;
```

(c) Assign the Role to the User Created in Lab 8(b)

Step 6: Connect as SYSTEM

```
CONNECT system/password@ORCLPDB;
```

Step 7: Grant the Role

```
GRANT role_by_suresh
```

```
TO user_suresh;
```

(d) View Roles Granted to a User

Step 8: Display Roles Granted to the User

```
SELECT GRANTEE,  
       GRANTED_ROLE  
FROM DBA_ROLE_PRIVS  
WHERE GRANTEE = 'USER_SURESH';
```

(e) Display Privileges Granted to the Role

Step 9: Display System Privileges

```
SELECT ROLE,  
       PRIVILEGE  
FROM ROLE_SYS_PRIVS  
WHERE ROLE = 'ROLE_BY_SURESH';
```

Step 10: Display Object Privileges

```
SELECT GRANTEE,  
       TABLE_NAME,  
       PRIVILEGE  
FROM DBA_TAB_PRIVS  
WHERE GRANTEE = 'ROLE_BY_SURESH';
```

(f) Create Another Role

Step 11: Create a New Role

```
CREATE ROLE manager_role;
```

Step 12: Grant Privileges

```
GRANT CREATE SESSION,  
       CREATE VIEW  
TO manager_role;
```

(g) Modify the Role Created in (f)

Step 13: Grant an Additional Privilege

```
GRANT CREATE SEQUENCE
```

```
TO manager_role;
```

Step 14: Revoke a Privilege

```
REVOKE CREATE VIEW
```

```
FROM manager_role;
```

(h) Drop the Role Created in (f)

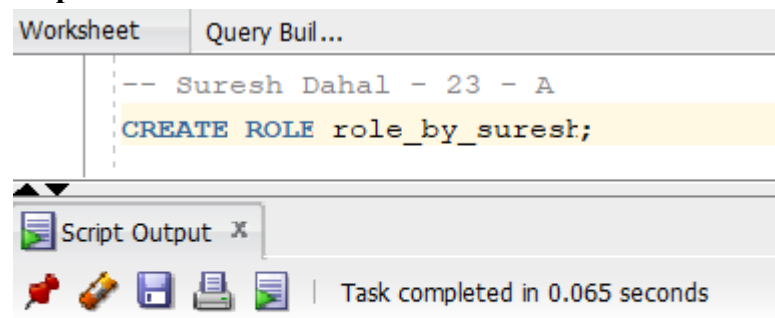
Step 15: Drop the Role

```
DROP ROLE manager_role;
```

OUTPUT:

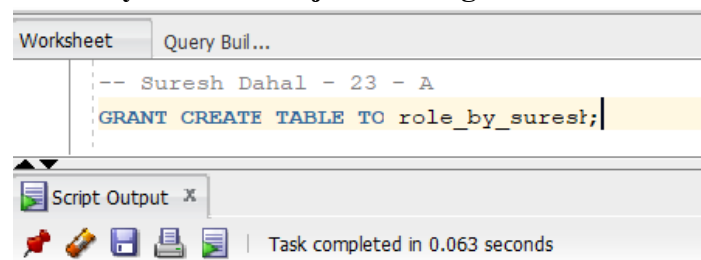
(a) Create a Role Named role_by_yourName

Step 1: Create the Role



```
Role ROLE_BY_SURESH created.
```

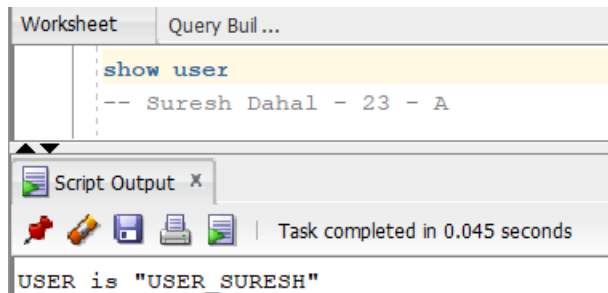
(b) Grant System and Object Privileges to the Role



```
Grant succeeded.
```

Step 2: Grant a System Privilege

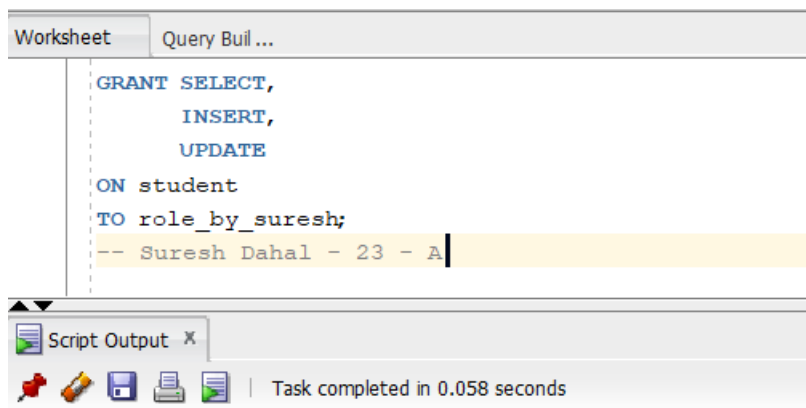
Step 3: Connect as the Owner of the Table



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a query: `show user`. Below the query, the output is shown: `-- Suresh Dahal - 23 - A`. The 'Script Output' window is open, showing the message: `USER is "USER_SURESH"`. The status bar indicates 'Task completed in 0.045 seconds'.

```
Worksheet    Query Buil ...
-- Suresh Dahal - 23 - A
show user
-- Suresh Dahal - 23 - A
Script Output X
Task completed in 0.045 seconds
USER is "USER_SURESH"
```

Step 5: Grant Object Privileges



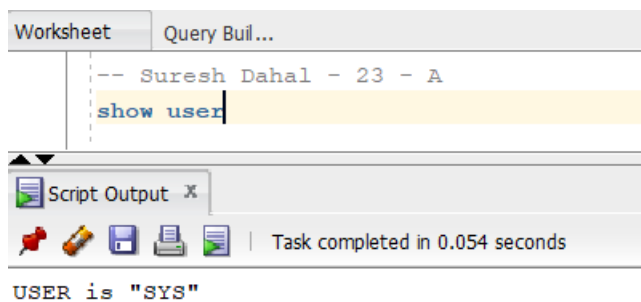
The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a GRANT statement: `GRANT SELECT, INSERT, UPDATE ON student TO role_by_suresh;`. Below the statement, the output is shown: `-- Suresh Dahal - 23 - A`. The 'Script Output' window is open, showing the message: `Grant succeeded.`. The status bar indicates 'Task completed in 0.058 seconds'.

```
Worksheet    Query Buil ...
GRANT SELECT,
      INSERT,
      UPDATE
ON student
TO role_by_suresh;
-- Suresh Dahal - 23 - A
Script Output X
Task completed in 0.058 seconds
Grant succeeded.
```

Grant succeeded.

(c) Assign the Role to the User Created in Lab 8(b)

Step 6: Connect as SYSTEM



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a query: `show user`. Below the query, the output is shown: `-- Suresh Dahal - 23 - A`. The 'Script Output' window is open, showing the message: `USER is "SYS"`. The status bar indicates 'Task completed in 0.054 seconds'.

```
Worksheet    Query Buil ...
-- Suresh Dahal - 23 - A
show user
-- Suresh Dahal - 23 - A
Script Output X
Task completed in 0.054 seconds
USER is "SYS"
```

Step 7: Grant the Role

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A
GRANT role_by_suresh
TO user_suresh;
```

Script Output x

Task completed in 0.053 seconds

Grant succeeded.

(d) View Roles Granted to a User

Step 8: Display Roles Granted to the User

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A
SELECT GRANTEE,
       GRANTED_ROLE
FROM DBA_ROLE_PRIVS
WHERE GRANTEE = 'USER_SURESH';
```

Script Output x

Task completed in 0.05 seconds

GRANTEE	GRANTED_ROLE
USER_SURESH	ROLE_BY_SURESH

(e) Display Privileges Granted to the Role

Step 9: Display System Privileges

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A
SELECT ROLE,
       PRIVILEGE
FROM ROLE_SYS_PRIVS
WHERE ROLE = 'ROLE_BY_SURESH';
```

Script Output x

Task completed in 0.072 seconds

ROLE	PRIVILEGE
ROLE_BY_SURESH	CREATE TABLE

Step 10: Display Object Privileges

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A ]
SELECT GRANTEE,
       TABLE_NAME,
       PRIVILEGE
FROM DBA_TAB_PRIVS
WHERE GRANTEE = 'ROLE_BY_SURESH';
```

Script Output x

Task completed in 0.058 seconds

GRANTEE	TABLE_NAME	PRIVILEGE
ROLE_BY_SURESH	STUDENT	INSERT
ROLE_BY_SURESH	STUDENT	SELECT
ROLE_BY_SURESH	STUDENT	UPDATE

(f) Create Another Role

Step 11: Create a New Role

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A ]
create role manager_role
```

Script Output x

Task completed in 0.045 seconds

Role MANAGER_ROLE created.

Step 12: Grant Privileges

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A ]
GRANT CREATE SESSION,
      CREATE VIEW
TO manager_role;
```

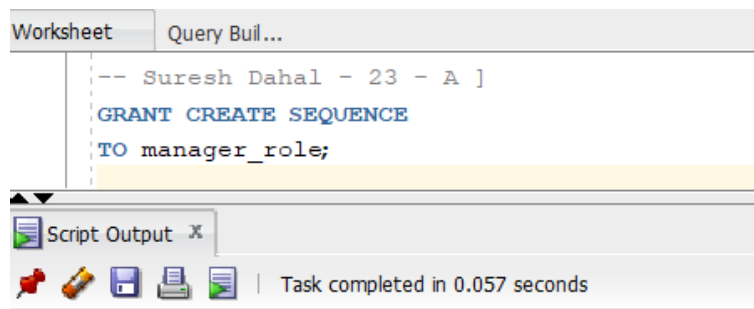
Script Output x

Task completed in 0.05 seconds

Grant succeeded.

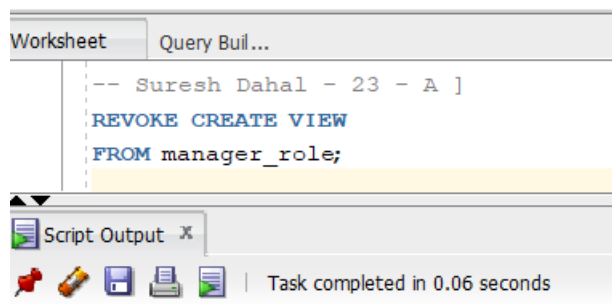
(g) Modify the Role Created in (f)

Step 13: Grant an Additional Privilege



Grant succeeded.

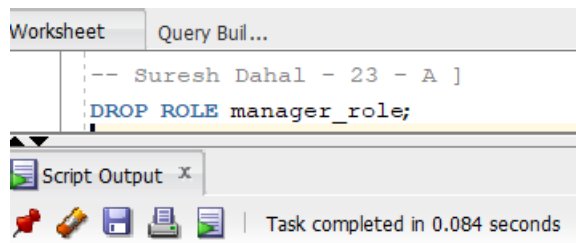
Step 14: Revoke a Privilege



Revoke succeeded.

(h) Drop the Role Created in (f)

Step 15: Drop the Role



Role MANAGER_ROLE dropped.

10. WORKING WITH PROFILES.

THEORY:

A **profile** is a collection of resource limits and password management policies assigned to database users. Profiles help database administrators control resource usage and enforce password security. A profile can be assigned to one or more users and managed using the CREATE PROFILE, ALTER PROFILE, DROP PROFILE, and ALTER USER statements.

STEPS:

(a) Create a Profile Named profile_by_yourName

Step 1: Connect as SYSTEM

Connect as system in Oracle SQL Developer.

Step 2: Create the Profile

```
CREATE PROFILE profile_by_suresh
```

```
LIMIT
```

```
SESSIONS_PER_USER 2
```

```
CPU_PER_SESSION 10000
```

```
CONNECT_TIME 60
```

```
IDLE_TIME 15
```

```
FAILED_LOGIN_ATTEMPTS 3
```

```
PASSWORD_LIFE_TIME 30
```

```
PASSWORD_GRACE_TIME 7;
```

(b) Assign the Profile to the User Created in Practical 8(b)

Step 3: Assign the Profile

```
ALTER USER user_suresh
```

```
PROFILE profile_by_suresh;
```

(c) View Profile Information

Step 4: Display All Profiles

```
SELECT PROFILE,
```

```
RESOURCE_NAME,
```

```
LIMIT  
FROM DBA_PROFILES  
ORDER BY PROFILE;
```

Step 5: Display Information About the Created Profile

```
SELECT PROFILE,  
        RESOURCE_NAME,  
        LIMIT  
FROM DBA_PROFILES  
WHERE PROFILE = 'PROFILE_BY_SURESH';
```

(d) Create Another Profile

Step 6: Create a New Profile

```
CREATE PROFILE test_profile  
LIMIT  
        SESSIONS_PER_USER 1  
        IDLE_TIME 10  
        PASSWORD_LIFE_TIME 20;
```

(e) Modify the Profile and Display Updated Information

Step 7: Modify the Profile

```
ALTER PROFILE test_profile  
LIMIT  
        IDLE_TIME 20  
        PASSWORD_LIFE_TIME 45;
```

Step 8: Display the Updated Profile

```
SELECT PROFILE,  
        RESOURCE_NAME,
```

```
LIMIT  
FROM DBA_PROFILES  
WHERE PROFILE = 'TEST_PROFILE';
```

(f) Drop the Profile and Display Updated Information

Step 9: Drop the Profile

```
DROP PROFILE test_profile;
```

If the profile is assigned to any user, use:

```
DROP PROFILE test_profile CASCADE;
```

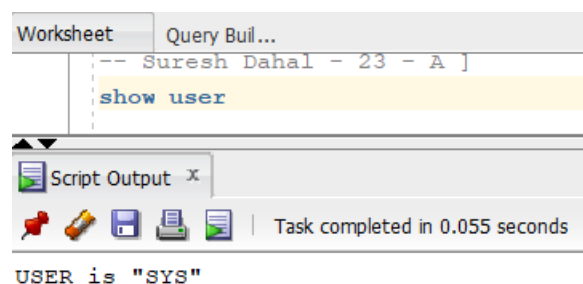
Step 10: Display the Remaining Profiles

```
SELECT PROFILE,  
       RESOURCE_NAME,  
       LIMIT  
FROM DBA_PROFILES  
ORDER BY PROFILE;
```

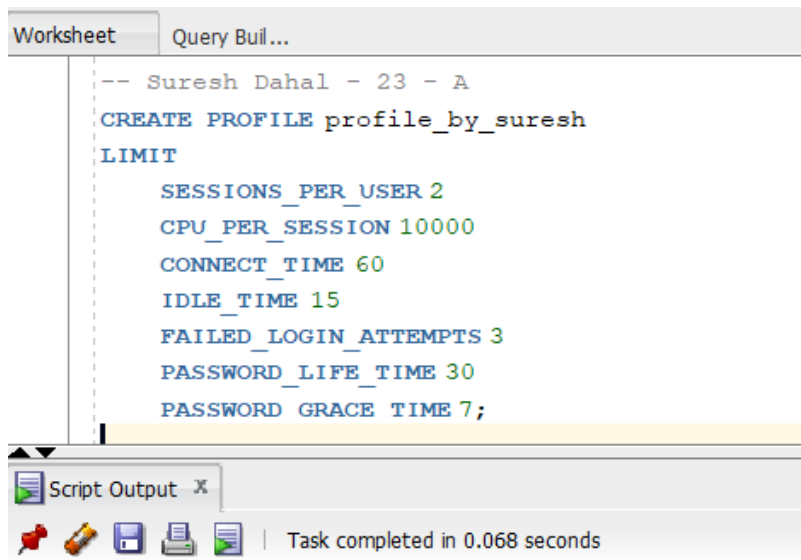
OUTPUT:

(a) Create a Profile Named profile_by_yourName

Step 1: Connect as SYSTEM



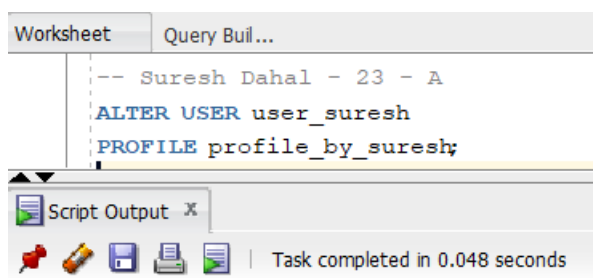
Step 2: Create the Profile



Profile PROFILE_BY_SURESH created.

(b) Assign the Profile to the User Created in Practical 8(b)

Step 3: Assign the Profile



User USER_SURESH altered.

(c) View Profile Information

Step 4: Display All Profiles

0.073 seconds

Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A
SELECT PROFILE,
       RESOURCE_NAME,
       LIMIT
FROM DBA_PROFILES
ORDER BY PROFILE;
```

Script Output x

Task completed in 0.073 seconds

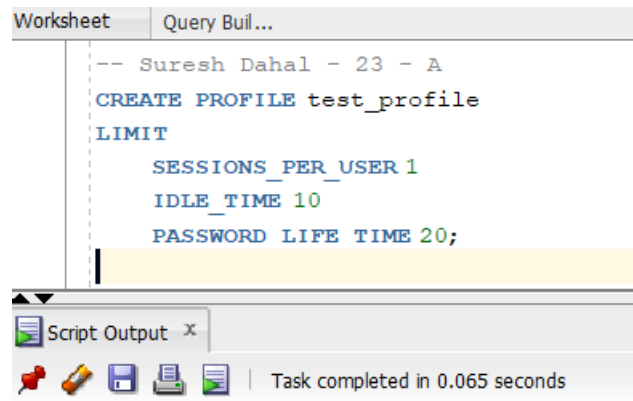
PROFILE	RESOURCE_NAME	LIMIT
DEFAULT	CONNECT_TIME	UNLIMITED
DEFAULT	INACTIVE_ACCOUNT_TIME	UNLIMITED
DEFAULT	LOGICAL_READS_PER_CALL	UNLIMITED
DEFAULT	LOGICAL_READS_PER_SESSION	UNLIMITED
DEFAULT	CPU_PER_CALL	UNLIMITED
DEFAULT	CPU_PER_SESSION	UNLIMITED
DEFAULT	SESSIONS_PER_USER	UNLIMITED
DEFAULT	COMPOSITE_LIMIT	UNLIMITED
DEFAULT	PRIVATE_SGA	UNLIMITED
DEFAULT	FAILED_LOGIN_ATTEMPTS	10
DEFAULT	PASSWORD_LIFE_TIME	180

Step 5: Display Information About the Created Profile

PROFILE	RESOURCE_NAME	LIMIT
PROFILE_BY_SURESH	COMPOSITE_LIMIT	DEFAULT
PROFILE_BY_SURESH	SESSIONS_PER_USER	2
PROFILE_BY_SURESH	CPU_PER_SESSION	10000
PROFILE_BY_SURESH	CPU_PER_CALL	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_SESSION	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_CALL	DEFAULT
PROFILE_BY_SURESH	IDLE_TIME	15
PROFILE_BY_SURESH	CONNECT_TIME	60
PROFILE_BY_SURESH	PRIVATE_SGA	DEFAULT
PROFILE_BY_SURESH	FAILED_LOGIN_ATTEMPTS	3
PROFILE_BY_SURESH	PASSWORD_LIFE_TIME	30
PROFILE	RESOURCE_NAME	LIMIT
PROFILE_BY_SURESH	PASSWORD_REUSE_TIME	DEFAULT
PROFILE_BY_SURESH	PASSWORD_REUSE_MAX	DEFAULT
PROFILE_BY_SURESH	PASSWORD_VERIFY_FUNCTION	DEFAULT
PROFILE_BY_SURESH	PASSWORD_LOCK_TIME	DEFAULT
PROFILE_BY_SURESH	PASSWORD_GRACE_TIME	7
PROFILE_BY_SURESH	INACTIVE_ACCOUNT_TIME	DEFAULT

(d) Create Another Profile

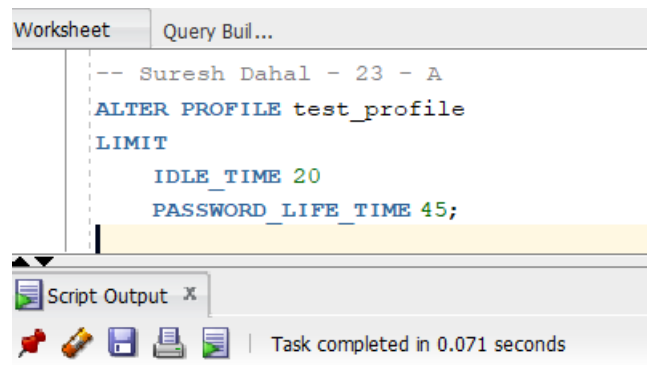
Step 6: Create a New Profile



Profile TEST_PROFILE created.

(e) Modify the Profile and Display Updated Information

Step 7: Modify the Profile



Profile TEST_PROFILE altered.

Step 8: Display the Updated Profile

Worksheet Query Buil...		
<pre>-- Suresh Dahal - 23 - A SELECT PROFILE, RESOURCE_NAME, LIMIT FROM DBA_PROFILES WHERE PROFILE = 'TEST_PROFILE';</pre>		
Script Output ✕      Task completed in 0.081 seconds		
PROFILE	RESOURCE_NAME	LIMIT

TEST_PROFILE	COMPOSITE_LIMIT	DEFAULT
TEST_PROFILE	SESSIONS_PER_USER	1
TEST_PROFILE	CPU_PER_SESSION	DEFAULT
TEST_PROFILE	CPU_PER_CALL	DEFAULT
TEST_PROFILE	LOGICAL_READS_PER_SESSION	DEFAULT
TEST_PROFILE	LOGICAL_READS_PER_CALL	DEFAULT
TEST_PROFILE	IDLE_TIME	20
TEST_PROFILE	CONNECT_TIME	DEFAULT
TEST_PROFILE	PRIVATE_SGA	DEFAULT
TEST_PROFILE	FAILED_LOGIN_ATTEMPTS	DEFAULT
TEST_PROFILE	PASSWORD_LIFE_TIME	45
PROFILE	RESOURCE_NAME	LIMIT

TEST_PROFILE	PASSWORD_REUSE_TIME	DEFAULT
TEST_PROFILE	PASSWORD_REUSE_MAX	DEFAULT
TEST_PROFILE	PASSWORD_VERIFY_FUNCTION	DEFAULT
TEST_PROFILE	PASSWORD_LOCK_TIME	DEFAULT
TEST_PROFILE	PASSWORD_GRACE_TIME	DEFAULT
TEST_PROFILE	INACTIVE_ACCOUNT_TIME	DEFAULT

(f) Drop the Profile and Display Updated Information

Step 9: Drop the Profile

Worksheet Query Buil...	
<pre>-- Suresh Dahal - 23 - A DROP PROFILE test_profile CASCADE;</pre>	
Script Output ✕      Task completed in 0.07 seconds	
Profile TEST_PROFILE dropped.	

Step 10: Display the Remaining Profiles

Worksheet

Query Buil...

-- Suresh Dahal - 23 - A

SELECT PROFILE,
RESOURCE_NAME,
LIMIT
FROM DBA_PROFILES
ORDER BY PROFILE;

Script Output .x

  | Task completed in 0.087 seconds

PROFILE	RESOURCE_NAME	LIMIT
ORA_STIG_PROFILE	FAILED_LOGIN_ATTEMPTS	3
PROFILE_BY_SURESH	COMPOSITE_LIMIT	DEFAULT
PROFILE_BY_SURESH	INACTIVE_ACCOUNT_TIME	DEFAULT
PROFILE_BY_SURESH	CPU_PER_SESSION	10000
PROFILE_BY_SURESH	CPU_PER_CALL	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_SESSION	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_CALL	DEFAULT
PROFILE_BY_SURESH	IDLE_TIME	15
PROFILE_BY_SURESH	CONNECT_TIME	60
PROFILE_BY_SURESH	PASSWORD_GRACE_TIME	7
PROFILE_BY_SURESH	PASSWORD_LOCK_TIME	DEFAULT